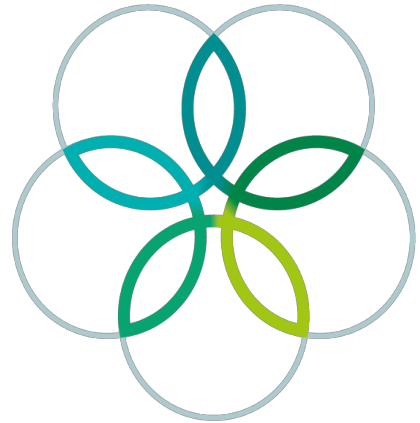
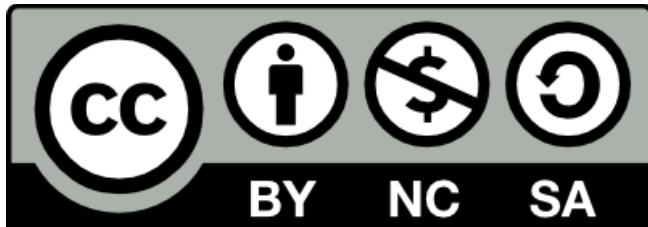


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30th International Biology Olympiad SZEGED, HUNGARY



Practical Exam 1.

Sugar chromatography

Starch identification

Microbial fuel cell

16th July 2019

COUNTRY

LANGUAGE

Practical exam 1.**General instructions**

This exam consists of three subtasks:

- Subtask 1. Analysis of sugars with thin layer chromatography (30 points)
- Subtask 2. Identification of a starch of unknown plant origin (30 points)
- Subtask 3. Microbial fuel cell (40 points)

1. **Please remember to attach your barcode sticker to all pieces of paper on the answer sheet.**
2. Write your answers in the separate answer sheet provided. **Only answers given in the answer sheet will be considered.**
3. Ensure you received all necessary materials and equipment listed on the next page. If any items are missing indicate this by raising your red card within 10 minutes following the start of the exam.
4. During experiments ensure all materials and equipment is handled properly. Any spilled solutions or broken equipment will not be replaced.
5. Stop answering and put down your pencil immediately when the bell rings signalling the end of the exam.
6. No paper, materials or equipment should be taken out of the laboratory.

CAUTION: The experiment deals with materials that are fragile and sharp. Exercise caution when handling them.

WEAR ALL SAFETY EQUIPMENT PROVIDED AT ALL TIMES OTHERWISE THE LAB ASSISTANT WILL ASK YOU TO LEAVE THE ROOM.

Materials and Equipment

- Eluent solutions (boric acid : acetonitrile, 3:7 mixture) in glass bottle
- Sugar solutions in Eppendorf tubes (G-glucose, M-maltose, S-sucrose, X-honey)
- Thin Layer Chromatogram (TLC)
- Pencil
- Pasteur pipettes
- Micropipette tips
- Screw-cap container
- Hot air blower (use it at the nearest shared table to detect the developed sugars on the chromatogram)
- Gloves
- Ruler
- Lugol solution (in dropper bottle)
- Microscopy slides
- Cover slips
- Knife

- Eppendorf tubes (to be labelled by the student A-D)
- Unknown starch sample (labelled as 'K')
- A sachet of dry yeast (labelled: YEAST)
- pH = 7.0 phosphate buffer (PB) – in a bottle for liquids
- 0.02 M K₃[Fe(CN)₆]-solution (Fe³⁺) dissolved in PB – in a bottle for liquids
- Plastic (blue) inoculation loop for mixing the yeast suspension
- Aqueous citric acid solution for the diaphragm – in 50 ml Falcon tube (labelled 'Citric Acid')
- A strip of filter paper for the diaphragm
- 2 x 250 ml plastic cups for assembling the galvanic cell
- Tweezers for soaking and moving the diaphragm
- Metal clips
- Multimeter with appropriate cables
- Distilled water
- Graphite sheet for electrode (2 pcs)
- distilled water; label: W
- beaker with ice
- beaker with room temperature water
- thermometer
- 37 °C dry thermostat (one shared by 4 competitors)
- Microcentrifuge tube labelled 'M': size ladder marker
- Microcentrifuge tube labelled 'C': control which is the PCR product of the original gene
- Microcentrifuge tube labelled 'P1': PCR product of the sample taken from P1 type cells
- Microcentrifuge tube labelled 'P2': PCR product of the sample taken from P2 type cells
- Microcentrifuge tube labelled 'E': restriction endonuclease (*Pst*I.) – held on ice!
- gel electrophoresis chamber
- electrophoresis gel

SUBTASK 1. ANALYSIS OF SUGARS WITH THIN LAYER CHROMATOGRAPHY (TLC)

Different sugars can be readily separated by thin layer chromatography (TLC) analysis. In this task, you have to determine the most abundant sugar component of honey. During the drying and eluting period, you can also start working on the other tasks.

Be careful with the hot air blower because it can cause severe burns!

Do not forget you are working with chemicals, always wear gloves!

Please remember to attach your barcode sticker to all pages of the answer sheet.

MATERIALS and TOOLS:

- eluent solution (boric acid : acetonitrile, 3:7 mixture) in a glass bottle
- sugar solutions in Eppendorf tubes (G-glucose, M-maltose, S-sucrose, X-honey)
- Thin Layer Chromatogram (TLC)
- Pencil
- Pasteur pipettes
- Micropipette tips
- Screw-cap container
- Hot air blower (use it at the nearest shared table to detect the developed sugars on the chromatogram)
- Gloves
- Ruler

The Eppendorf tubes in the tube rack on your table contain the following samples:

G – glucose

M - maltose

S – sucrose

X – honey

The bottle labeled as 'ELUENT' contains a 3:7 mixture of boric acid solution and acetonitrile.

1. Put on gloves for the exercise!
2. Draw a straight line (START LINE) on the white side of the aluminium backed thin layer with a pencil parallel to the shorter side of the sheet, at about 1 cm from its end. Do not push the pencil too hard because otherwise, it will scratch off the thin layer!
3. Pipette with a Pasteur pipette a quantity of ELUENT solution in the screw-cap container up to just below the start line, BUT do not place the TLC plate in it, only verify from the outside that the liquid level is adequate. Screw on the cap of the container and wait 10 minutes for an equilibrium between the eluent and the gas phase above.
4. While waiting, use the Pasteur pipette with a micropipette tip to apply 1 mm of diameter spots of each sugar solution. The spots should be just above the START LINE, with 1 cm far from each other. Use a new pipette tip for each sample.
5. Leave the plate prepared this way on the table.
6. After visible drying, wait 5 more minutes before proceeding.
7. Place the TLC plate with the uploaded samples in the container and screw the cap on.

8. Continue eluting until the eluent reaches 1 cm distance before the end of the plate (it takes about 10-15 minutes). Then remove the plate from the screw-cap container.
9. Mark the solvent (eluent) front by drawing a line on the TLC plate with a pencil.
10. Go to the closest free desk with the shared devices. Set the hot air blower to Level II and dry the TLC plate by placing it on the tile with its white surface upward and blowing it with hot air, until you can see the brown spots, then turn off the hot air blower. Be careful, the TLC is very hot for the next couple of minutes, do not touch it!

Be careful with the hot air blower because it can cause severe burns!

Q.1.1.1 When finished (the TLC has cooled down), put your chromatogram into the envelope labeled "Chromatogram". **Write the 3-digit number on the envelope in the code box on the Answer Sheet. When you are done, CALL THE LAB ASSISTANT AND SUBMIT YOUR ENVELOPE! (This action is necessary to proceed to the next questions.)**

When you submit your own chromatogram image, the lab assistant will give you a new one required to complete the next tasks. (This chromatogram was run in the same manner as the one performed earlier.

Based on the provided chromatogram image, perform the following measurements and then calculate the retention factors (R_f). Use the widest part of each spot for measuring the distance.

The distance of solvent front ('ELUENT') from the 'START' line: _____ cm

$$R_f = \frac{d}{E}$$

Where:

d: distance of the spot from the START line

E: distance of ELUENT front from the START line

Q.1.1.2 Fill the TABLE 1 on the answer sheet.

Q.1.1.3 Based on the chromatogram given, identify the carbohydrate(s) present in the honey sample. Put an X under the labels (G, S, M) of the appropriate sugars.

SUBTASK 2. IDENTIFICATION OF A STARCH OF AN UNKNOWN PLANT ORIGIN

In this practical your task will be to study and draw several types of starch grains.

1. Using your knife scrap the surface of your biological samples onto separated microscopic slide each. (A: bean, B: wheat grain, C: potato, D: rice grain) Write the appropriate letter (A, B, C or D) on the slide with the marker.
2. Drip one drop of Lugol's solution (iodine solution with KI) on the scrapings, cover it with a cover slip and examine them at a 100x total magnification under a microscope.

Q.1.2.1 (No marks allocated for this point, nevertheless if your sample is not approved it means that you can't get more marks in this task.)
When all the slides are prepared, adjust a scraping of your choice to sharp resolution and call the lab supervisors to prove your work. Be aware that you need these slides for the next tasks.

Q.1.2.2 Draw at least two starch grains per each sample (A-D). Make sure you draw each starch grain in sufficient detail, they should reflect the shape, pattern, and (relative) size of each grain.

You have received a food starch sample, used in the food industry as a thickener, in an Eppendorf tube (labelled as 'K'). Test a very small portion of this, by diluting with drops of water and Lugol's solution.

Q.1.2.3 Based on your observations from the previous task, determine from which plant the sample came. Indicate your answer by putting an X in the appropriate box(es) on your answer sheet. Note that putting an X into any inappropriate box causes loss of half of the total marks for Q.1.2.3. until you reach zero.

Q.1.2.4 Assume that the average volume of a starch grain is $4000 (\mu\text{m})^3$. The starch grain can be considered a 20% aqueous solution of starch with a density of 1.050 g/cm^3 . What is the mass of the starch constituting this one grain? Give you answer in ng.

Q.1.2.5 How many glucose subunits does the above amount of starch grain consist of? Assume that the grain consists of amylose only, with an average number of 1000 subunits per molecule. Give the result with 3 significant figures in scientific notation. The molar mass of glucose is 180 g/mol ; that of water is 18 g/mol ; Avogadro number (NA): $6 \cdot 10^{23} \text{ 1/mol}$.

SUBTASK 3. MICROBIAL FUEL CELL

DURING THIS PRACTICAL YOU WILL WORK WITH CARBON TISSUE. BE AWARE THAT IT IS EXTREMELY DANGEROUS TO HANDLE IT WITHOUT GLOVES!

Methylene blue is a widely used dye, also used as a redox indicator in addition to dyeing microscopic preparations and macromolecules. Since this material is not toxic at a low concentration, bacterial and yeast cells can absorb it without suffering significant damage. However, due to its redox properties and good diffusion abilities, it is known that at an appropriate concentration is able to intervene in redox processes in cells - "tapping" electrons from the latter. For example, its ability to gain electrons from glucose, thus converted to gluconic acid, is well known. In this task we try to discover whether it is capable of achieving this with a material participating in the oxidative phosphorylation (terminal oxidation) process in the mitochondria. Hungarian researchers published a paper in 2017 (Komlódi, Tretter; 2017), outlining a possible model of a mechanism in which methylene blue can help neurons get ATP in some neurodegenerative diseases (e. g. Alzheimer's disease).

Your task is now to test the electrode potential (mid-point potential) of methylene blue, thus confirming or rebutting that methylene blue is indeed capable to participate in the above process in this way. You will make your decision on the basis of the following chemical principle: an agent with the more negative redox potential is able to reduce (transfer an electron to) the agent with the more positive redox potential.

The mid-point potential is the electrode potential value measured when a reduced and oxidized form of a redox agent is present in equal amounts. In the reaction you perform, this type of equilibrium is reached by a galvanic cell and yeast cells. Living cells continuously reduce methylene blue, the reduced form of methylene blue is released from the cells, where they will form the anode of the galvanic cells, continuously supply the appropriate electrode with electrons. Oxidation, electron transfer of methylene blue is also continuous on the electrode, thus the ratio of the reduced and oxidized forms is constant in both the cells and the fluid surrounding the electrode, the mid-point potential can be determined as the

solution's electrode potential.

Even at this time, the galvanic cell will consist of two electrodes: methylene blue release an electron on one electrode (anode) and $K_3[Fe(CN)_6]$ (Fe^{3+}) takes up an electron on the other one (cathode) and is thus converted to $K_4[Fe(CN)_6]$ containing ferrous (Fe^{2+}) ions. The galvanic cell is kept "alive" while oxidized methylene blue molecules are able to enter the yeast cells where they are reduced in the above-mentioned redox reaction, so the galvanic cell is not depleted as long as there are live yeast cells and enough Fe^{3+} ions.

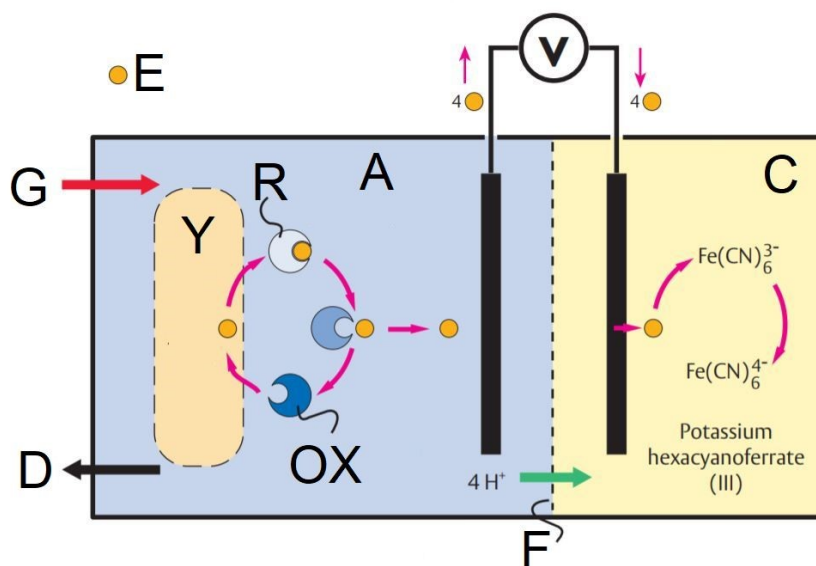


Fig.1. Legend: A: anode. C: Cathode. D: CO_2 . E: electron. F: Diaphragm connecting the two electrolytes. G: Glucose. OX: oxidized form of methylene blue. R: Reduced form of methylene blue. Y: Yeast cell.

Your task is to assemble this live galvanic cell and to measure its electromotive force. After you showed the results of your measurement, you will receive information on a separated sheet to be used at the calculation tasks.

Assembly of a galvanic cell:

**To measure volumes use the scaling on the wall of the cups (in mL-s).
You should use the two scaled colourless plastic cups in the following procedure. Put Both into the white rectangular box before you start to create the fluid of the anode and the cathode.**

When preparing the anode fluid, follow the steps below:

1. Pour about 20 mL of phosphate buffer (PB) to a 250 mL cup and label it as 'ANODE'
2. Add the total amount of the provided powdered yeast to the buffer.
3. Stir the mixture with the inoculation loop until a suspension with uniform texture (homogeneous) is formed.
4. Add about 20 mL of glucose solution (G) to the mixture obtained thereby further homogenizing the yeast suspension.
5. Then add 20 mL of methylene blue solution (M) to the mixture thereby further homogenizing the suspension.

When preparing the cathode fluid, follow the steps below:

1. Pour 60 mL of yellow $K_3[Fe(CN)_6]$ solution (Fe^{3+}) into the other glass vessel.

When assembling the diaphragm, follow the steps below:

The diaphragm is intended to provide electrical connection between the two solutions (i. e. it has to reach down into both solutions.) This function will be fulfilled by a filter paper strip soaked in citric acid solution in this case by means of highly mobile hydrogen ions in the solution.

1. Put the cathode fluid on the right and the anode fluid on the left.
2. Put the two cups of your forming galvanic cell next to each other so that their edges come into contact.
3. Fold the filter paper in half along its shorter side (i. e. making it half as long as before).
4. Then, completely dip the prepared filter paper strip into the citric acid solution by using the tweezers and let it soak for ca. 5 seconds.
5. Lift out the moistened filter paper strip, let it drain off, then place it in the two plastic cups, so that the two tips hanging down into each of the solutions.
6. This is important, because the filter paper should not come into contact with the graphite electrodes as this would cause a short circuit.

Assembly of a galvanic cell – measurement of the electromotive force:

1. Now, only the electrodes should be placed in both solutions prepared this way, then connected to the voltage meter. To arrange the voltaic cell you can use all the tools provided.
2. IF YOU HAVE FINISHED UP TO THIS POINT, RAISE YOUR **YELLOW CARD**, AND HAVE YOUR ASSEMBLED SYSTEM CHECKED BY YOUR LAB ASSISTANT, WHO WILL SIGN YOUR ANSWER SHEET (Q1.3.1)
3. Take off the caps from both the black and the red cables of the multimeter.
4. Take the multimeter's black cable into your left hand and the multimeter itself into your right hand.
5. Switch on the multimeter with turning the black wheel on it to the 'V' sign.
6. Touch the two carbon electrode with the two needle points and read the voltage.

Q.1.3.2 Record the measured voltage on the Answer Sheet.

After you recorded your results a scotch tape will be stacked on your Answer Sheet. You also receive a Voltage value with which you have to calculate in the following tasks.

Calculation tasks for the microbial fuel cell

You received a voltage value from the lab assistant you have to calculate with!

The following are known:

- The standard redox potential of the $[\text{Fe}(\text{CN})_6]^{3-} + e^- = \text{K}_3[\text{Fe}(\text{CN})_6]^{4-}$ reaction: 0.361 V
- Molar mass of $\text{K}_3[\text{Fe}(\text{CN})_6]$: 329.24 g/mol
- Molar mass of methylene blue: 319.85 g/mol
- The $\text{K}_3[\text{Fe}(\text{CN})_6]$ -solution was prepared by dissolving 3.39 g of salt in 500 mL of PB.
- The methylene blue solution was prepared by dissolving 1.87 g of dye in 500 mL of PB.
- Calculation of electrode potentials of solutions with an approximate concentration of $1 \frac{\text{mol}}{\text{dm}^3}$ is specified by the Nernst equation:

$$\varepsilon = \varepsilon_0 + \frac{RT}{zF} \ln c$$

Where:

- ϵ : the cell potential
- ϵ^0 : the standard redox potential
- R : the universal gas constant (8.314 J/mol K)
- T : the absolute temperature
- F : the molar charge (96486 C/mol)
- z : the number of moving electrons (methylene blue also loses/gains four electrons per round)
- c : concentration expressed in mol/L

A measurement performed under optimal circumstances was completed with the following results:

- The temperature was 25 °C
- 10% of the methylene blue was present in the anode solution – the rest was in the yeast cells.
- Electromotive Force (EMF) is the voltage you have measured.

Q.1.3.3 What is the value of the RT/F multiplication factor for both electrodes (calculated in SI units)? Give your answer to 5 decimal places on the answer sheet.

Q.1.3.4 What is the electrode potential of the cathode solution in this setup? Give your answer in millivolts rounded to an integer.

Q.1.3.5 What is the electrode potential of the anode solution in this setup? Give your answer in millivolts as an integer.

Q.1.3.6 Enter the concentration of methylene blue in the anode solution. Give your answer in mol/L rounded to 2 significant figures.

Q.1.3.7 Enter the standard redox potential of methylene blue based on the measurement data. Give your answer in millivolts as an integer.

Based on another calculation, the midpoint potential of methylene blue was 11 mV. The table below shows the standard redox potentials of biological oxidation processes. Based on the midpoint potential value of methylene blue and the figures in the table below, write X in the appropriate cell(s) on the answer sheet.

Methylene Blue is capable of ...

A. taking up electrons from

B. transferring electrons to

Q.1.3.8 NADH

Q.1.3.9 Cytochrome C

Q.1.3.10 Succinate

Electrode equation	E'_0 (V)
Succinate + CO₂ + 2 H⁺ + 2 e⁻ ⇌ α-ketoglutarate + H₂O	-0.670
Acetate + 2 H⁺ + 2 e⁻ ⇌ acetaldehyde	-0.580
2 H⁺ + 2 e⁻ ⇌ H₂	-0.421
α-Ketoglutarate + CO₂ + 2 H⁺ + 2 e⁻ ⇌ isocitrate	-0.380
Cystine + 2 H⁺ + 2 e⁻ ⇌ 2 cysteine	-0.340
NAD⁺ + 2 H⁺ + 2 e⁻ ⇌ NADH + H⁺	-0.320
NADP⁺ + 2 H⁺ + 2 e⁻ ⇌ NADPH + H⁺	-0.324
Acetaldehyde + 2 H⁺ + 2 e⁻ ⇌ ethanol	-0.197
Pyruvate + 2 H⁺ + 2 e⁻ ⇌ lactate	-0.185
Oxaloacetate + 2 H⁺ + 2 e⁻ ⇌ malate	-0.166
FAD + 2 H⁺ + 2 e⁻ ⇌ FADH₂ (in flavoproteins)	+0.031
Fumarate + 2 H⁺ + 2 e⁻ ⇌ succinate	+0.031
Ubiquinone + 2 H⁺ + 2 e⁻ ⇌ ubiquinol	+0.045
2 cytochrome <i>b</i>_(ox) + 2 e⁻ ⇌ 2 cytochrome <i>b</i>_(red)	+0.070
2 cytochrome <i>c</i>_(ox) + 2 e⁻ ⇌ 2 cytochrome <i>c</i>_(red)	+0.254
2 cytochrome <i>a</i>_{3(ox)} + 2 e⁻ ⇌ 2 cytochrome <i>a</i>_{3(red)}	+0.385
$\frac{1}{2}$ O₂ + 2 H⁺ + 2 e⁻ ⇌ H₂O	+0.816

Table 1. Standard Redox Potentials of Selected Half-Reactions

30th International Biology Olympiad SZEGED, HUNGARY



Practical Exam 2.

Blood Sugar Measurement

Molecular Biology

16th July 2019

COUNTRY

LANGUAGE

Practical Exam 2.

General instructions

This exam consists of two subtasks:

- Subtask 1. A model of home blood glucose measurement (50 points)
- Subtask 2. Examination of knock-out mutant cell lines produced by CRISPR-Cas 9 method (50 points)

1. **Please remember to attach your barcode sticker to all pieces of paper on the answer sheet.**
2. Write your answers in the separate answer sheet provided. **Only answers given in the answer sheet will be considered.**
3. Ensure you received all necessary materials and equipment listed on the next page. If any items are missing indicate this by raising your red card within 10 minutes following the start of the exam.
4. During experiments ensure all materials and equipment is handled properly. Any spilled solutions or broken equipment will not be replaced.
5. Stop answering and put down your pencil immediately when the bell rings signalling the end of the exam.
6. No paper, materials or equipment should be taken out of the laboratory.

CAUTION: The experiment deals with materials that are fragile and sharp. Exercise caution when handling them.

WEAR ALL SAFETY EQUIPMENT PROVIDED AT ALL TIMES OTHERWISE THE LAB ASSISTANT WILL ASK YOU TO LEAVE THE ROOM.

Materials and Equipment

- Eppendorf tubes
- 10 μ L micropipette
- 20 μ L micropipette
- 200 μ L micropipette
- 1000 μ L micropipette
- micropipette tips
- permanent marker
- calculator
- red signaling card
- yellow signaling card
- tissues
- ACCU-CHEK Active blood glucose meter
- Test strip container with 15 test strips – keep closed to prevent damage to the strips
- standard glucose solution in distilled water; label: G; 1 mL
- distilled water; label: W
- beaker with ice

- beaker with room temperature water
- thermometer
- 37 °C dry thermostat (one shared by 4 competitors)
- Microcentrifuge tube labelled 'M': size ladder marker
- Microcentrifuge tube labelled 'C': control which is the PCR product of the original gene
- Microcentrifuge tube labelled 'P1': PCR product of the sample taken from P1 type cells
- Microcentrifuge tube labelled 'P2': PCR product of the sample taken from P2 type cells
- Microcentrifuge tube labelled 'E': restriction endonuclease (*Pst*I.) – held on ice!
- gel electrophoresis chamber
- electrophoresis gel

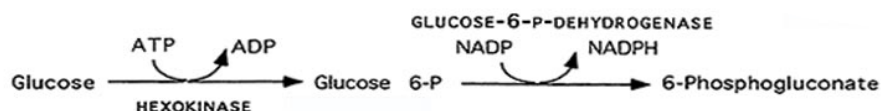
SUBTASK 1. A MODEL OF HOME BLOOD GLUCOSE MEASUREMENT

Glucose in the mammalian body is metabolized in a complex system which provides a balance (homeostasis) of highly regulated processes.

Sufficient levels of blood glucose are maintained hormonally and allosterically by the regulation of transporters and key enzymes of glucose metabolism.

Metabolic diseases that result in a disturbance of these regulating mechanisms have severe consequences to the affected individuals and have become of economic significance on a global level. The most common of them all – diabetes mellitus – is a group of metabolic diseases that have chronic hyperglycemia (elevated blood glucose levels) in common as a result of disturbances in insulin secretion and/or the effects of insulin. For the management of diabetes, monitoring blood glucose levels is essential. Especially patients undergoing insulin therapy are required to measure their blood glucose levels on a daily basis, since the calculation of the correct doses of insulin is essential to avoid hypo- and hyperglycemia, both life-threatening situations. Correct insulin dosage and consistently sufficient blood glucose levels are also key elements for prognosis improvement.

In this practical exam, handheld glucose meters developed for home-use by diabetes patients will be used to simulate blood glucose measurement in model glucose solutions. Glucometers are used for home blood glucose monitoring by patients with diabetes mellitus and other diseases with a high risk for hypoglycemia. The glucometer used for these determines blood glucose content with reflectance photometry using the hexokinase method:

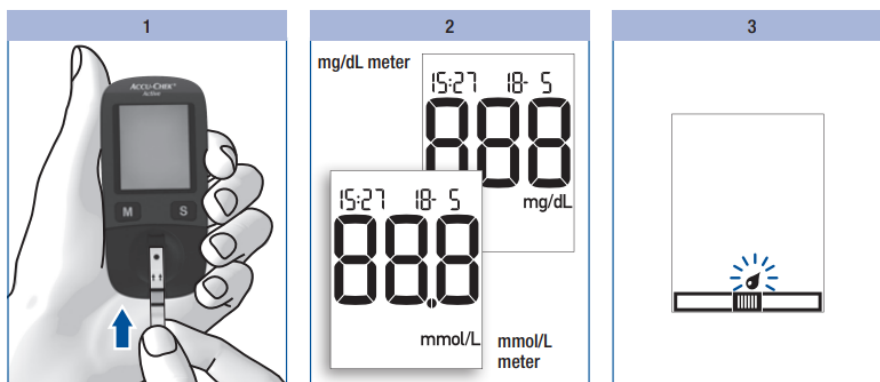


Since glucometers are calibrated using whole blood, calibration curves have to be prepared to obtain correct measurement results when using aqueous solutions.

How to Measure Glucose Concentrations with the ACCU-CHEK Active Blood Glucose Meter (see figure below)

1. Take a test strip from the test strip container. Close the container immediately.
2. Hold the test strip so that the arrows printed on it and the green square face upwards
3. Slide the test strip into the test strip guide in the direction of the arrows until you feel it lock into place. (1) Do not bend the test strip.
4. The test strip must lie flat on the measurement window cover.
5. The meter turns on and first performs a default display check (approx. 2 seconds). (2)
6. Following the display check, the test strip symbol and the flashing blood drop symbol appear on the display (3).
7. The meter is now ready to perform a blood glucose test. You have approx. 90 seconds to apply 3 μL of glucose solution onto the middle of the green square of the test strip. After this time, the meter turns itself off. If this happens, remove the strip and slide it in again.
8. The test starts. The flashing hourglass symbol indicates that the test is in progress.
9. The test is over in approximately 5 seconds.
10. You can now mark the test result.
11. After removing the test strip, the device will turn itself off. Discard any used test strips.

(If an error appears before the application of the liquid, remove the test strip and try again. If it doesn't work the second time or you already applied the glucose solution, discard the test strip and use a new one.)



Part 1. Glucometer Measurement of the Concentration of Model Glucose Solutions

First, you are going to be testing the accuracy of glucose measurement with the glucose meter on water-based model glucose solutions.

Procedure

1. Prepare the following dilutions from the provided standard glucose solution in Eppendorf tubes using distilled water: 1:5; 1:2; 1:1; 2:1 (in a sample volume : diluent volume format) – 300 μL of each. Label the tubes accordingly.
2. Measure the glucose content of the solutions.

Q.2.1.1 Fill in the table on the answer sheet.

Table Legend:

D – Dilution (standard volume : diluent volume)

V_S – added volume of standard solution

V_{H_2O} – added volume of distilled water

MC – Measured glucose concentration, one decimal place accuracy

Dependence of Blood Glucose Measurement on Sample Temperature

Glucose concentration values measured by some glucometers are affected by temperature. To confirm or refute this, you will measure the glucose concentration of the solution previously diluted 1:1 at different temperatures.

Procedure

1. Label 4 Eppendorf tubes with the respective temperature. You are going to measure the glucose concentrations at 0 °C, 10 °C, room temperature (RT) and 37 °C. **On the tube for 37 °C, also add your competitor's ID number legibly.**
2. To each tube, add 50 µL of the 1:1 diluted glucose solution.
3. Move the solutions to the respective temperatures: 0 °C in ice water, 10-15 °C by cooling room temperature water with ice and controlling the temperature with a thermometer, room temperature left on the rack and 37 °C in the dry thermostat. Leave the solutions to equilibrate for at least 5 minutes. You'll have to pay attention to keep the 10-15 °C beaker at the right temperature range.
4. After at least 5 minutes, measure the glucose concentrations with the glucometer. You have to work quickly at this point, not to let the samples differ too much from the required temperature.

Q.2.1.2 Record the results of your measurements in the table on the answer sheet.

Table Legend: T – solution temperature RT – room temperature MC – Measured glucose concentration, one decimal place accuracy

When you're finished, **signal with yellow card**. For Part 2, an assistant will hand you the results of related experiments, which you'll have to analyze. Your own measurement results will be taped over with translucent tape so **beyond this point, no more changes on part 1 answers will be possible**.

Part 2. Preparation of a Calibration Curve Based on Pre-Acquired Results

Presented are the results of several glucometer measurements of glucose concentrations in aqueous glucose solutions analogous to the measurements you carried out in Part 1

SC [mmol/L]	MC [mmol/L]
4.0	4.5
6.0	6.4
8.3	8.0
9.9	9.6
13.6	13.3
16.0	15.0

Legend: SC – standard glucose concentration. MC – glucose concentration measured by glucometer

Q.2.1.3 Plot the measured concentrations (y) against the standard concentration (x) in the graph on the answer sheet as a scatter plot. Use small but defined dots, less than 2 mm of diameter. Pay attention to the correct labeling of the axes; use the abbreviations in the table above.

Q.2.1.4 Applying simple linear regression, calculate the regression equation for the best fitting line using the least square method. Calculate at 2 decimals accuracy. Use the equation below. Make sure to include the calculated values in the table on the answer sheet. The grey columns are provided for convenient calculation and will not be marked.

Table Legend: SC – standard glucose concentration MC – glucose concentration measured by glucometer n – number of measurement

Q.2.1.5 Draw your calculated line in the graph according to the completed formula below.

Regression equation: $y = a + bx$

$$s_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

$$s_y = \sqrt{\frac{\sum (y - \bar{y})^2}{n - 1}}$$

$$r = \frac{((x - \bar{x})(y - \bar{y}))}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

$$b = r \frac{s_y}{s_x}$$

$$a = \bar{y} - b\bar{x}$$

Where n = number of measurements

Comparison of Blood Glucometer Readings at Different Temperatures

S	L15	R30
1	5.4	6.7
2	4.2	5.2
3	6.0	6.8
4	4.4	6.2
5	5.7	9.4
6	4.6	5.5
7	5.1	8.7
8	5.9	8.6
9	3.7	2.6
10	4.4	4.4
11	4.8	7.0
12	5.6	8.9
13	3.9	5.7
14	4.3	6.4
15	4.5	5.4

Legend: n – subject number. L15 – glucose meter reading at 15 °C. R30 – glucose meter reading at 30 °C

Q.2.1.6 On the answer sheet, fill in the table with values of 2 decimals precision (the grey columns are provided for convenient calculation and will not be marked).

Table Legend:

D – the difference (L15 - R30) for each test subject;

n – number of test subjects

Q.2.1.7 Perform a paired Student's t test to determine if there's a difference between the readings at different temperatures at a significance level of $\alpha = 0.001$. Indicate your calculated t value (t_{stat}) and critical t value (t_{crit}) according to the t value distribution table, with 2 decimals precision. (See formula below)

Decide whether the following statements are true (T) or false (F) considering your calculations. Place one X per statement in one of the boxes in the table on the answer sheet.

Q.2.1.8 The null hypothesis assumes that the true mean difference between the 15 °C and 30 °C measurements is zero.

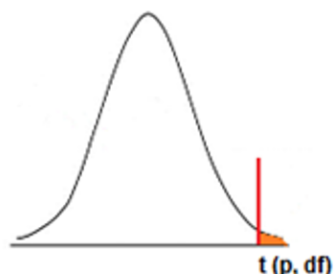
Q.2.1.9 The value of t_{stat} is lower than the value of t_{crit} , therefore the null hypothesis cannot be rejected.

$$t = \frac{\sum D}{\sqrt{\frac{n \sum D^2 - (\sum D)^2}{n-1}}}$$

Legend: D – the difference (L15 - R30) for each test subject; n – number of test subjects; tstat – t value of the statistics; tcrit – critical value of t according to the t distribution table

Right-sided t-distribution

Legend: df – degrees of freedom



df/p	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.0005
1	0.324920	1.000000	3.077684	6.313752	12.70620	31.82052	63.65674	636.6192
2	0.288675	0.816497	1.885618	2.919986	4.30265	6.96456	9.92484	31.5991
3	0.276671	0.764892	1.637744	2.353363	3.18245	4.54070	5.84091	12.9240
4	0.270722	0.740697	1.533206	2.131847	2.77645	3.74695	4.60409	8.6103
5	0.267181	0.726687	1.475884	2.015048	2.57058	3.36493	4.03214	6.8688
6	0.264835	0.717558	1.439756	1.943180	2.44691	3.14267	3.70743	5.9588
7	0.263167	0.711142	1.414924	1.894579	2.36462	2.99795	3.49948	5.4079
8	0.261921	0.706387	1.396815	1.859548	2.30600	2.89646	3.35539	5.0413
9	0.260955	0.702722	1.383029	1.833113	2.26216	2.82144	3.24984	4.7809
10	0.260185	0.699812	1.372184	1.812461	2.22814	2.76377	3.16927	4.5869
11	0.259556	0.697445	1.363430	1.795885	2.20099	2.71808	3.10581	4.4370
12	0.259033	0.695483	1.356217	1.782288	2.17881	2.68100	3.05454	4.3178
13	0.258591	0.693829	1.350171	1.770933	2.16037	2.65031	3.01228	4.2208
14	0.258213	0.692417	1.345030	1.761310	2.14479	2.62449	2.97684	4.1405
15	0.257885	0.691197	1.340606	1.753050	2.13145	2.60248	2.94671	4.0728
16	0.257599	0.690132	1.336757	1.745884	2.11991	2.58349	2.92078	4.0150
17	0.257347	0.689195	1.333379	1.739607	2.10982	2.56693	2.89823	3.9651
18	0.257123	0.688364	1.330391	1.734064	2.10092	2.55238	2.87844	3.9216
19	0.256923	0.687621	1.327728	1.729133	2.09302	2.53948	2.86093	3.8834
20	0.256743	0.686954	1.325341	1.724718	2.08596	2.52798	2.84534	3.8495
21	0.256580	0.686352	1.323188	1.720743	2.07961	2.51765	2.83136	3.8193
22	0.256432	0.685805	1.321237	1.717144	2.07387	2.50832	2.81876	3.7921
23	0.256297	0.685306	1.319460	1.713872	2.06866	2.49987	2.80734	3.7676
24	0.256173	0.684850	1.317836	1.710882	2.06390	2.49216	2.79694	3.7454
25	0.256060	0.684430	1.316345	1.708141	2.05954	2.48511	2.78744	3.7251
26	0.255955	0.684043	1.314972	1.705618	2.05553	2.47863	2.77871	3.7066
27	0.255858	0.683685	1.313703	1.703288	2.05183	2.47266	2.77068	3.6896
28	0.255768	0.683353	1.312527	1.701131	2.04841	2.46714	2.76326	3.6739
29	0.255684	0.683044	1.311434	1.699127	2.04523	2.46202	2.75639	3.6594
30	0.255605	0.682756	1.310415	1.697261	2.04227	2.45726	2.75000	3.6460
∞	0.253347	0.674490	1.281552	1.644854	1.95996	2.32635	2.57583	3.2905

SUBTASK 2. EXAMINATION OF KNOCK-OUT MUTANT CELL LINES PRODUCED BY CRISPR-CAS9 METHOD

In this practical task you will examine constructs created with a modern gene-editing procedure, the CRISPR-Cas9 technique. With this technique, we can 'knock out' (KO) genes much more efficiently and quickly relative to the previously used homologous recombination technique. This made it easier than ever to investigate the effect of the lack of a gene selected by us. To facilitate the workflow even further, an antibiotic resistance gene may also be used which is incorporated into our target gene. This not only renders the target gene non-functional, but also provides an easy means of selecting successfully mutated cells by adding the appropriate antibiotic to the growth medium.

Part 1. The theory of PCR-based gene insertion detection

The following task describes the testing of CRISPR-induced mutant cell lines using a PCR and restriction digest based method. Assume that all of the experiments are carried out on diploid mammalian cells.

Using our CRISPR system, we insert the antibiotic resistance gene (PURO) that codes for an enzyme cleaving the antibiotic puromycin. Due to the characteristics of this method, the PURO gene may be inserted both ways. In many cases – not discussed in detail here – the orientation of insertion has a significant impact on the outcome of the entire experiment, so as a rule, the orientation of insertion should be checked. In order to achieve a complete KO we should also determine whether the resistance gene has been inserted into only one or both chromosomes, since a non-functional KO mutant can only be expected in the latter case. Last but not least, it should also be checked that the PURO gene is inserted into the desired target area or if it is at a different site. Some of these different scenarios are illustrated in Fig. 1. Please study this figure before proceeding.

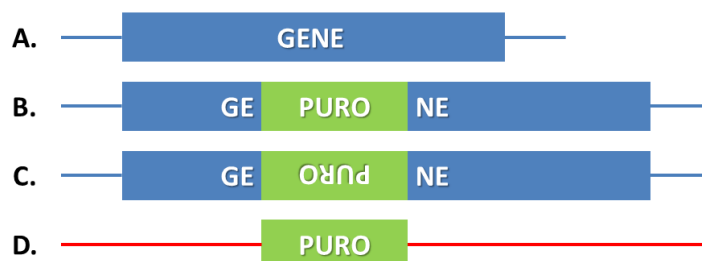


Fig. 1. A.: target gene; B.: PURO insertion in forward orientation; C.: PURO insertion in reverse orientation; D.: PURO insertion into the genome at a site other than the target gene.

To examine the above insertions, a PCR reaction is designed using three kind of primers. We design primers to match both ends of the original gene, and a third primer to match the end of the PURO gene opposite the promoter region (i.e., it is complementary to the 3' end). Then we examine the products obtained in a normal PCR reaction. It is a well-known phenomenon that in the case of competitive primers appearing during a PCR reaction, the amplification of shorter DNA region is expected to be predominant. In the following simplified gel images the less intense bands are not indicated.

Q.2.2.1 Match the lanes (1-5) of the gel image of PCR products (Fig. 2) with the schematics of potential genomic insertions (Fig. 3, A-F) with an **X** in your answer sheet.

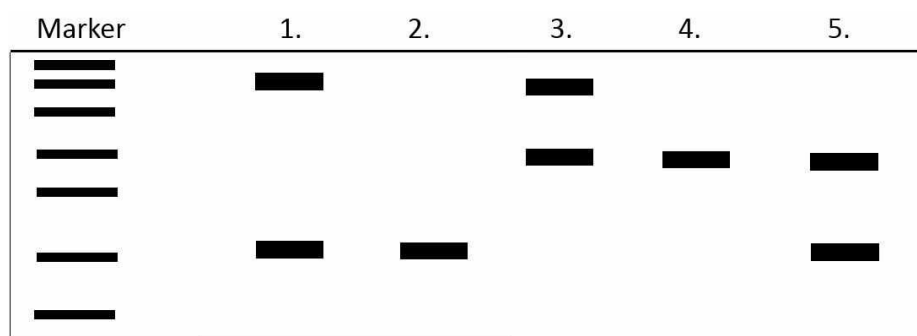


Fig. 2. Image of gel electrophoresis of PCR products

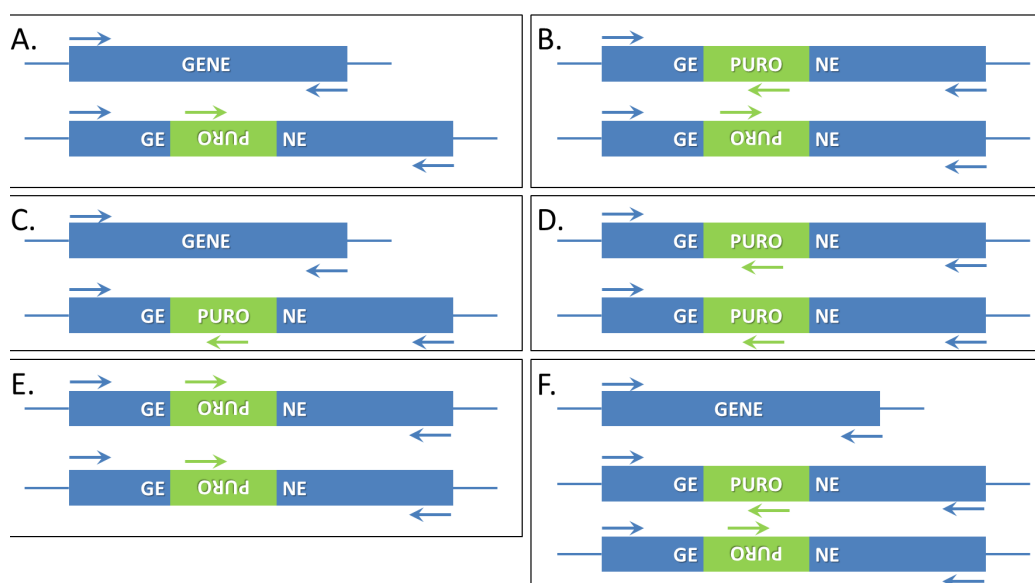


Fig. 3. Possible insertion scenarios

Now that you understand how to check in general the asymmetrical insertion characteristics of the Puro gene in diploid cells using a PCR reaction, you may face another issue in relation to research work. Sometimes it occurs that – again for reasons not described here in detail – the Puro gene has to be inserted in the middle region of the target gene. Suppose that the 500-bp long Puro gene is to be inserted into the centre of a 1400-bp gene. It is obvious that in this case the orientation of insertion cannot be detected by the above simple 3-primer PCR reaction, because the sizes of the resulting products are the same, regardless of the direction of insertion.

Considering that in this size range, a difference of 100 bp is hardly visible on gels with such resolution, the above statement can be extended to cases where insertion takes place near the centre of the target.

In some fortunate cases, however, both the target gene and the Puro gene already contain a restriction enzyme cleavage site in an asymmetrical position. In this case, after the restriction cleavage of the PCR product prepared using 2 primers complementary to the ends of the target gene, the orientation of insertion can still be determined. Study Fig. 4 in the given model system.

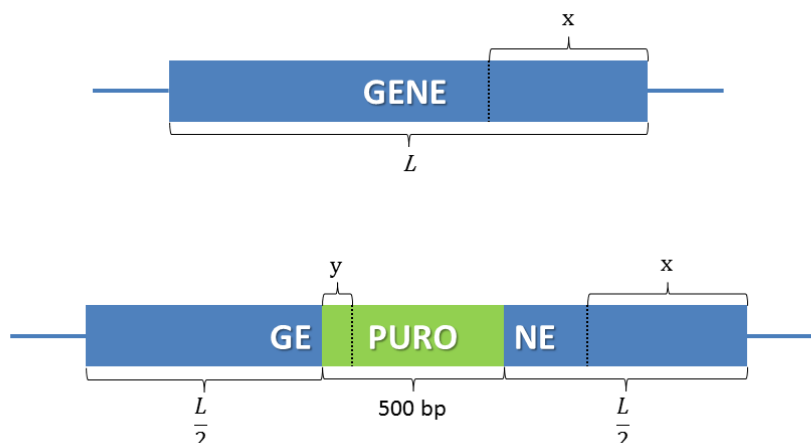


Fig. 4 Schematic of the restriction based insertion mapping. L = length of the target gene in base pairs, x = distance of the restriction site from the end in the target gene (in bp), y = distance of the restriction site from the start of the PURO gene.

	L	x	y
A	1400	500	100
B	1000	500	100
C	1400		
D	1000		

Table 1. L = Length of the target gene in base pairs (bp). x = distance of the restriction site from the end in the target gene (bp), y = distance of the restriction site from the start of the PURO gene (bp).

In Table 1 you can see different symmetrical insertion scenarios as shown in Fig. 4. Suppose that the products are amplified with primers complementing the ends of the target gene and then fully cleaved with restriction enzyme E.

Q.2.2.2 A - How many DNA bands are visible during electrophoresis after enzymatic digestion when the PURO gene is inserted in the same orientation as the target gene in Case A Table 1.?

B - How many DNA bands are visible during electrophoresis after enzymatic digestion when the PURO gene is inserted in the opposite orientation as the target gene in Case B Table 1.?

C - How many bp is the size of the longest DNA band after enzymatic digestion when the PURO gene is inserted in the opposite orientation as the target gene in Case C Table 1.?

D - How many bp is the size of the shortest DNA band after enzymatic digestion when the PURO gene is inserted in the opposite orientation as the target gene in Case D Table 1.?

Q.2.2.3 Indicate the letter of the scenario (A-D, Table 1.) with an **X** in which not even the restriction digestion method helps to determine the orientation of insertion.

Part 2. Restriction mapping of gene insertion

Now that you understand how this procedure works, you will examine an actual gene insertion process. In the current practical, we attempted to insert the PURO gene with a length of 500 bp in the centre of a 1000-bp long target gene (for clarity's sake, precisely after the 500th base pair) using the CRISPR technique. It is also known that the value of X also shown in Fig. 4 is 300 bp, whereas the value of Y is 100 bp. The researchers carried out the insertion by CRISPR and selected five genetically stable, puromycin-resistant cell lines (P1 to P5). Subsequently, DNA from each cell line was isolated, and the DNA region of interest was amplified using a primer pair complementary to both ends of the target gene.

Of the five experimental products, you will examine two: a completely non-functional KO mutant (P1) and a mutant with partial loss of function (P2), in which RNA of normal length is transcribed from the target gene at a rate of approx. 50% in the cells. The goal of your examination is to carry out enzyme restriction cleavage with the obtained products and to separate the digestion products by gel electrophoresis.

Procedure:

IMPORTANT! The samples for the DNA analysis are provided in coloured microcentrifuge tubes, in a storage box.

- Microcentrifuge tube labelled 'M': size ladder marker
- Microcentrifuge tube labelled 'C': control which is the PCR product of the original gene
- Microcentrifuge tube labelled 'P1': PCR product of the sample taken from P1 type cells
- Microcentrifuge tube labelled 'P2': PCR product of the sample taken from P2 type cells
- Microcentrifuge tube labelled 'E': restriction endonuclease (*Pst*I.) – held on ice!

1. The DNA samples already contain the loading dye, which does not interfere with the enzyme activity.
2. Add 2 μ l of 'E' to tubes 'P1' and 'P2' using the 2–20 μ l micropipette respectively. Set the pipette to 18 μ l and mix the contents of the tube by aspirating and releasing it with the pipette.
3. Clearly label your tube with the abbreviation of the name of country.
4. Place the tubes in the designated dry thermostat (intended for shared use), which is already set at the digestion temperature of 37°C, for at least 15 minutes (no need to be accurate to the second). The accuracy of the wall clock in the room is sufficient.
5. At the end of the incubation period, remove tubes 'P1' and 'P2' from the dry thermostat and place them on ice for approx. 1 minute, then put them back in the tube storage box.
6. Remove the orange lid of the electrophoresis equipment. Check the buffer level in the equipment. The buffer should cover the gel completely. If this is not the case, pour some buffer into the gel tank. You can ask the lab assistant for help by raising your **red card**.
7. Transfer 18 μ l into each well of the gel as follows:

Well 1: M

Well 2: C

Well 3: P1

Well 4: P2

8. Put the orange lid back on the electrophoresis equipment and inform the lab assistant by raising your **yellow card**.

Q.2.2.4 Your lab assistant will note the time you started your gel (START). Do not write in that field.

9. Get the gel running with the help of the lab assistant. You can turn off the equipment with the power switch.
10. During the run, you can check the DNA bands by turning on the lights.
11. Run the gel for 20 minutes, gels run for longer or shorter times will get partial marks. When finished, call for the lab assistant immediately who will take a picture of your gel.
12. Signal to the lab assistant by raising your **yellow card**.

Q.2.2.4 Your lab assistant will note the time at the end of your gel run (END) and photograph your gel. Your gel running time (TIME) will be calculated based on your start and end times. Do not write in these fields.

Q.2.2.5 Enter the three-digit number located on the gel tray on the answer sheet (CODE). Your score will be identified based on this.

If you have entered the number on the sheet and the picture has also been taken, you will receive a **virtual gel image** from the lab assistant. **Use this image to answer the following question.** Lane C is the control, non-digested sample. Lanes (P1-P5) are five puromycin resistant cell lines.

Q.2.2.6 Indicate with an **X** in your answer sheet all type(s) of PURO insertions, **as seen in Fig. 1** (A-D), found in samples P1-P5 on the gel image you received.

30th International Biology Olympiad
SZEGED, HUNGARY



Practical Exam 3.

Plant Anatomy

Ecology

Animal Anatomy

16th July 2019

COUNTRY

LANGUAGE

Practical Exam 3.

General instructions

This exam consists of three subtasks:

- Subtask 1. Plant Identification (30 points)
- Subtask 2. Composition of Grasslands (20 points)
- Subtask 3. Examination of a chicken's neck and wing

1. **Please remember to attach your barcode sticker to all pieces of paper on the answer sheet.**
2. Write your answers in the separate answer sheet provided. **Only answers given in the answer sheet will be considered.**
3. Ensure you received all necessary materials and equipment listed on the next page. If any items are missing indicate this by raising your red card within 10 minutes following the start of the exam.
4. During experiments ensure all materials and equipment is handled properly. Any spilled solutions or broken equipment will not be replaced.
5. Stop answering and put down your pencil immediately when the bell rings signalling the end of the exam.
6. No paper, materials or equipment should be taken out of the laboratory.

CAUTION: The experiment deals with materials that are fragile and sharp. Exercise caution when handling them.

WEAR ALL SAFETY EQUIPMENT PROVIDED AT ALL TIMES OTHERWISE THE LAB ASSISTANT WILL ASK YOU TO LEAVE THE ROOM.

Materials and equipment

- Two unknown grass samples labelled A and B in zip-lock bags
- Magnifier with built in light
- Small scissors
- Dissecting pin
- 3 pcs longer pins (to affixing organs if required)
- Small tweezers
- Large tweezers
- 2 pcs of chicken neck and 2 pcs of chicken wings (washed)
- BOTTLE: small sealable bottle **with code marking** (filled with physiological saline solution)
- SACHET: small sealable sachet **with code marking**
- 1 small bottle of distilled water
- 1 squirt bottle

SUBTASK 1. PLANT IDENTIFICATION

Members of the grass family (*Poaceae*) have compound inflorescences, the base units of which are the *spikelets* (**Fig 1**). Spikelets can be considered simple inflorescences themselves, usually giving place to more flowers, but there are one-flowered spikelets as well. On the axis (**a**) of the spikelet, a *palea* can be found below the flowers, one on each side (**b**, **c**), but there are cases when only one palea can be found. Flowers (**d1**, **d2**) are located between the two paleas, on the spikelet axis. All flowers are very simplified, the perianth in most of the cases consists of only two *lemmas*. Lemmas are distinguished based on their position, one is called outer (or lower) lemma (**e**), the other is called inner (or upper) lemma (**f**). The outer lemma can be found on the spikelet axis, while the inner lemma is in the flower itself, on the flower axis (**i**). The outer lemma often has an awn (slender, bristle-like extension, **g**), which can be glabrous (hairless), rough or hairy. In the centre of each flower there is the ovary, on top of it with the stigma (**h**) that is adapted to air pollination, and is often feathery. The stamens (**j**) arise from below the ovary. There are lodicules (**k**) below the stamens.

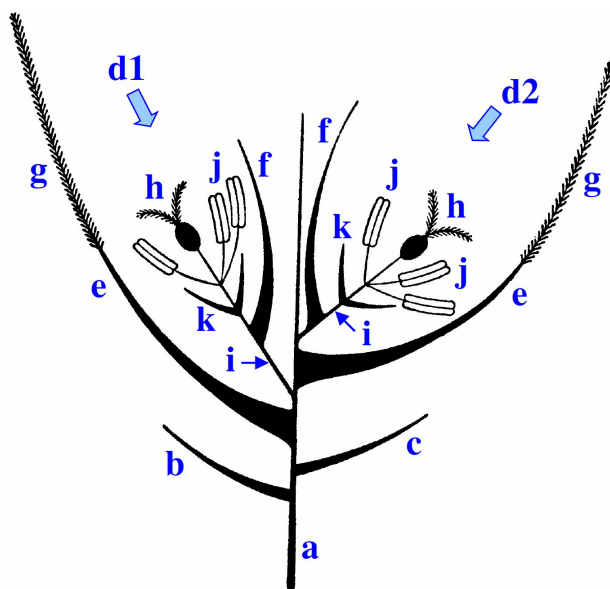


Fig.1 Schematic figure of a two-flowered spikelet

Q.3.1.1 Identify the two grass species (A and B) in the vials provided using the dichotomous key below. You may remove them from the bags for easier observation. Indicate the steps you take in the flow diagram in your answer sheet by writing the number of steps in the correct order. Mark any unused cells in the flow diagram with an **X**.

1. a. Spikelets form a raceme (an indeterminate inflorescence in which the main axis produces a series of flowers on lateral stalks, the oldest at the base and the youngest at the top), they are sitting directly on the main inflorescence axis or connected to it by a short (1-3 mm), not branching pedicel. **Go to 2.**
 - b. Spikelets are connected to the main inflorescence axis by branching pedicels. If pedicels are short, the inflorescence is a contracted panicle, if pedicels are long, spikelet form an open panicle (loose branching cluster of flowers). **Go to 7.**
2. a. There is only one spike on the stem apex. **Go to 3.**
 - b. The inflorescence is a cluster of spikes, there are more (usually 3-7) spikes terminally on the stem, starting from the same height. Individual spikes are 3-6 cm long. Small, approximately 2 mm long, violet, 1-2-flowered spikelets stand individually on one side of the spike axis. The stem is

prostrate-erect, 10-30 cm tall, the plant has stolons (slender, prostrate or trailing stems, producing roots and sometimes erect shoots at their nodes). The leaves are fibrous, 5-15 cm long, 2-4 mm wide, with a rough edge. **Species I.**

3. a. The spikelet has 2 paleas. **Go to 4.**
 b. The spikelet has only 1 palea. 10-20 mm long spikelets connect to the spike axis with their edge, sitting on alternating sides of the axis, therefore the spike is pronouncedly flattened. There are 6-10 flowers in a spikelet and 16-40 spikelets in a spike. The stem is cylindrical, erect, loosely spreading or upright in curves. **Species II.**
4. a. Spikelets stand alone on the axis side seats (ridge along the length of the axis). **Go to 5.**
 b. Spikelets stand two by two or three by three on the axis side seats. The spike is wide on the top and is narrowing to the base, 5-10 cm long, the axis is fragile, falling apart at maturation. The palea of mid-spikelets has long hairs. The lemma is 10 mm long at most, its awn is 20-30 mm long. The plant is bushy, the leaf blade is slightly hairy. **Species III.**
5. a. Spikelets are sessile (attached directly to the base). **Go to 6.**
 b. Spikelets are connected to the main inflorescence axis by 1-3 mm long pedicels. The spike inflorescence is nodding. Spikelets have 6-15 flowers, slightly compressed, their wider side turns towards inflorescence axis. The lemma is long awned, the awns are at least the same length as the lemma. Leaves are soft, lower ones are hairy. **Species IV.**
6. a. The spike inflorescence is flat, 2-6 cm long, 1-2.3 cm wide. Spikelets stand densely, at 1 mm from each other, they are ascending, 8-15 mm long, having 3-5-flowers. The palea has 1-3 veins, the lemma is short awned. Mat-forming plant, the stem is 25-60 cm long. **Species V.**
 b. The spike inflorescence is at least 8 cm long. Spikelets lie on the axis, the distance between them is at least 2 mm, they are 10-20 mm long, having 5-7-flowers. The palea has 3-7 veins, the lemma is awned or not. The plant has long stolons, the stem is 40-150 cm long. **Species VI.**
7. a. Spikelets are one-flowered. **Go to 8.**
 b. Spikelets consist of 2 or more flowers. **Go to 9.**
8. a. The outer lemma is 15-17 mm long, its awn is 25-33 mm long. The awn above the lemma is hairy, feathery. Spikelets stand loose on the flower axis. Outer and inner lemmas grow onto the caryopsis (a dry, indehiscent, one-seeded fruit in which the seed coat is closely fused to the fruit wall). The leaf is 0.5-1 mm in diameter, usually rolled, sometimes flat. **Species VII.**
 b. On the back of the outer lemma, a geniculate (bent at a right angle) awn rises, extending over the lemma, protruding 2-7 mm from the spikelet. The contracted spike-like panicle (loose branching cluster of flowers) is evenly cylindrical, 3-9 cm long, 8-10 mm wide. Paleas are joined to almost their mid-points or above. Leaf blade is flat, 3-10 mm wide. **Species VIII.**
9. a. The spikelet axis is, at least under the lower flower, lying on the lemma base, with short hairs. **Go to 10.**
 b. The spikelet axis is totally glabrous. **Go to 11.**
10. a. The spikelet is 2-flowered, approximately 10 mm long. One of the flowers is small and infertile, the other is mature and awned. The inflorescence is a narrow panicle (loose branching cluster of flowers), 10-30 cm long. The lemma is 7-9 mm long, its awn arises in the lower third of the back side, the awn of lower flowers is 10-20 mm long. The plant is 100-150 cm tall, slightly mat-forming. Leaf blade is 20-40 cm long, 5-10 mm wide. **Species IX.**
 b. The spikelet is 2-4-flowered, 15-30 mm long. The inflorescence is a 15-30 cm long loose panicle, spikelets are located on long, nodding lateral branches. The plant is 60-150 cm tall, bushy. Leaf blade is 30 cm long, 10-16 mm wide. **Species X.**
11. a. The spikelets are heart-shaped, round or triangular, 4-7 mm long and wide, laterally slightly compressed, 4-12-flowered, nodding. The lemma is blunt tipped, without awn. The inflorescence is a loose panicle (loose branching cluster of flowers), 8-20 cm long. The plant is 15-75 cm tall. Leaf blade is 4-16 cm long, 1.9-3.2 mm wide. **Species XI.**

b. The spikelets are narrower, 3-4 mm wide. The lemma is 2.6-3.4 mm long, its awn is 0.8-1.2 mm long. The inflorescence is a panicle, 2-5 cm long. The plant grows in dense clusters, is small, 10-30 cm tall. The leaf blade is folded along the main vein, its diameter is 0.4-0.6 mm. ***Species XII.***

Q.3.1.2 Indicate the identity of the two grass species (A and B) in your answer sheet with an **X**.

SUBTASK 2. COMPOSITION OF GRASSLANDS

Xerophilous loess grasslands are Central Europe's typical plant associations, rich in species. Their stock-forming plant species originate mainly from drought-tolerant grasses. Their grazed and slightly degraded communities are characterized by *Festuca pseudovina*, *Stipa capillata* and *Cynodon dactylon*. In this task, you will determine the relative abundance of three grass species and any association between them, based on an actual field survey. During field the sampling, we have observed the species occurring in 500 successive quadrats along a line. The size of quadrats were 10 x 10 cm.

Q.3.2.1 Complete the three contingency tables in your answer sheet with the mutual occurrences (a - d as well as the appropriate row and column sums) of the three different pairs of species in the first 40 quadrats (Table 1). Letters entered in each cell (a , b , c , d) serve to denote the variables of mutual occurrences (a : number of quadrats where both species were present, b and c : number of quadrats where only one or the other species was present, d : number of quadrats in which none of the examined species was present, + present, - not present). The variable n is the number of sampling units (quadrats), which is 40 in this case. All present species have been surveyed of which the following table only shows the occurrence of the three tested grass species sequentially in the first 40 quadrats. The value of 1 indicates the presence and 0 the absence of the given species.

<i>Festuca pseudovina</i>	00000 00000 00000 00000 00101 00000 00001 11011
<i>Stipa capillata</i>	01100 11101 11010 10000 11110 01111 10110 00010
<i>Cynodon dactylon</i>	11000 00000 00010 00000 01000 00000 00000 00000

Table 1. Occurrence data from the first 40 quadrants

Conclusions on the relationship of the spatial pattern of species were drawn from the occurrence of species in the quadrats in the entire sample (500 quadrats). The relationship was analysed using a χ^2 -test (chi-square test) referring to a positive or negative dependence or independence of occurrences.

The contingency tables for all quadrats ($n = 500$) sampled during the survey for the three pairs of species are as follows:

		<i>Festuca</i>		
		+	-	
<i>Stipa</i>	+	$a = 22$	$b = 271$	$a+b = 293$
	-	$c = 90$	$d = 117$	$c+d = 207$
		$a+c = 112$	$b+d = 388$	$n = 500$

		<i>Festuca</i>		
		+	-	
<i>Cynodon</i>	+	$a = 130$	$b = 163$	$a+b = 293$
	-	$c = 24$	$d = 183$	$c+d = 207$
		$a+c = 154$	$b+d = 346$	$n = 500$

		<i>Cynodon</i>		
		+	-	
<i>Stipa</i>	+	$a = 14$	$b = 140$	$a+b = 154$
	-	$c = 98$	$d = 248$	$c+d = 346$
		$a+c = 112$	$b+d = 388$	$n = 500$

Q.3.2.2 Determine the number of degrees of freedom (df) required for your χ^2 -test in a four field contingency table and write it in the appropriate cell in your answer sheet.

Q.3.2.3 Based on the formula provided, calculate the χ^2 -value for the last contingency table rounded to **two decimal places** and note it in your answer sheet. The first two χ^2 -values have already been filled in for you.

$$\chi^2 = \frac{(n-1)(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

Equation 1. The χ^2 formula, where the individual letters refer to the labelled fields (a-d) in the contingency table.

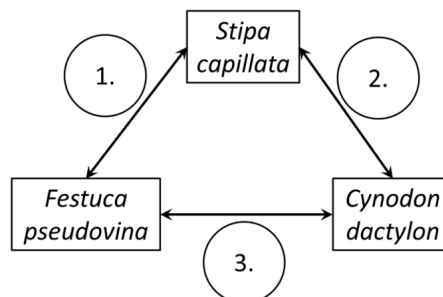
Q.3.2.4 Indicate the level of statistical significance (p-value) based on the χ^2 table (Table 2.) by writing one of the letters (A-D) in the appropriate cells in your answer sheet.

	<i>p-value</i>			
	A	B	C	D
<i>df</i>	$p > 0.05$	$0.05 \geq p > 0.01$	$0.01 \geq p > 0.001$	$0.001 \geq p$
1	$\chi^2 < 3.841$	$3.841 \leq \chi^2 < 6.635$	$6.635 \leq \chi^2 < 10.83$	$10.83 \leq \chi^2$
2	$\chi^2 < 5.991$	$5.991 \leq \chi^2 < 9.210$	$9.210 \leq \chi^2 < 13.82$	$13.82 \leq \chi^2$
3	$\chi^2 < 7.815$	$7.815 \leq \chi^2 < 11.35$	$11.35 \leq \chi^2 < 16.27$	$16.27 \leq \chi^2$
4	$\chi^2 < 9.488$	$9.488 \leq \chi^2 < 13.28$	$13.28 \leq \chi^2 < 18.47$	$18.47 \leq \chi^2$
5	$\chi^2 < 11.07$	$11.07 \leq \chi^2 < 15.09$	$15.09 \leq \chi^2 < 20.52$	$20.52 \leq \chi^2$
6	$\chi^2 < 12.59$	$12.59 \leq \chi^2 < 16.81$	$16.81 \leq \chi^2 < 22.46$	$22.46 \leq \chi^2$
7	$\chi^2 < 14.07$	$14.07 \leq \chi^2 < 18.48$	$18.48 \leq \chi^2 < 24.32$	$24.32 \leq \chi^2$
8	$\chi^2 < 15.51$	$15.51 \leq \chi^2 < 20.09$	$20.09 \leq \chi^2 < 26.13$	$26.13 \leq \chi^2$
9	$\chi^2 < 16.92$	$16.92 \leq \chi^2 < 21.67$	$21.67 \leq \chi^2 < 27.88$	$27.88 \leq \chi^2$
10	$\chi^2 < 18.31$	$18.31 \leq \chi^2 < 23.21$	$23.21 \leq \chi^2 < 29.59$	$29.59 \leq \chi^2$

Table 2. Critical values of χ^2 -distribution at various degrees of freedom (df)

Q.3.2.5 Based on the above, evaluate the spatial relationships (1-3) between the three grass species with a X in your answer sheet.

- A. Positive relationship
- B. Independent
- C. Negative relationship



SUBTASK 3. EXAMINATION OF A CHICKEN'S NECK AND WING

Please note that all the animal tissues provided are obtained from chickens bred and slaughtered for food consumption.

Health and Safety Notices

- **Always pay attention to the correct usage of the tools. Work cautiously.**
- **Wear rubber gloves.**
- **Do not place your hand or fingers in front of the tip or the cutting edge of the scissors or the dissecting pins.**
- **When cutting or scraping, always move the tip or edge of dissecting tools away from your body.**
- **If you get injured, report it immediately!**

Materials and tools

- 2 pieces of chicken neck and 2 pieces of chicken wings
- Small scissors, dissecting pin, small forceps
- BOTTLE: small sealable bottle **with code marking** (filled with physiological saline solution)
- SACHET: small sealable sachet **with code marking**
- magnifier (with lighting)
- longer pins (3 pcs, to fix the organs if required)
- spraying flask (for moistening the organs during dissection, if required)

Contents

1. Dissection and observation of a chicken neck (Q3.3.1-5) - 21 MARKS
2. Dissection and observation of chicken wings (Q3.3.6-14) - 29 MARKS

Recommendations

- Use the descriptions of anatomical names and directions on the following two pages.
- Read the questions carefully before starting the dissection.
- Always consider innervation, position, origin, insertion, and function of a muscle.

INDICATE ALL YOUR ANSWERS BY WRITING AN "X" IN THE APPROPRIATE BOX ON THE ANSWER SHEET!

PUT ALL STRUCTURES THAT HAVE TO BE SUBMITTED IN THE CONTAINERS SPECIFIED AND LEAVE THEM ON YOUR TRAY AT THE END OF THE PRACTICAL!

Anatomical definitions and explanations

- Median sagittal plane: a longitudinal plane in the midline that divides the body/organ into a left and a right side. Also serves as a reference line to define medial and lateral:
 - Medial: positioned nearer to the central plane of the body.
 - Lateral: positioned further from the central plane of the body
- Transverse plane: it divides the body or the given structure into a frontal (anterior) and a back (posterior) part. The part in front of this plane (positioned nearer to the skull) is the cranial, while the one behind this plane (positioned nearer to the tail) is the caudal part.
- Flexion: the articulating bones move closer to each other, the angle between them is decreasing
- Extension: the articulating bones move away from each other, the angle between them is increasing
- Rotation: the bones or part of a bone turn along their longitudinal axis in the joint
- Proximal: the part positioned nearer to the trunk along the main longitudinal axis of the limbs
- Distal: the part positioned further away from the trunk along the main longitudinal axis of the limbs
- Dorsal: at the upper side or back
- Ventral: at the underside or abdomen
- Cranial: towards the head, cephalic
- Caudal: towards the tail
- Antagonist muscles: muscles that move in an opposite way compared to each other
- Synergist muscles: muscles that act in the same way, participating in moving an articulation to a certain direction.
- Tubercle: a small nodule
- Epicondyle: protuberance
- Origin of a muscle: the proximal point of attachment
- Insertion of a muscle: the distal point of attachment

Anatomical directions and planes in birds (Figure 1)

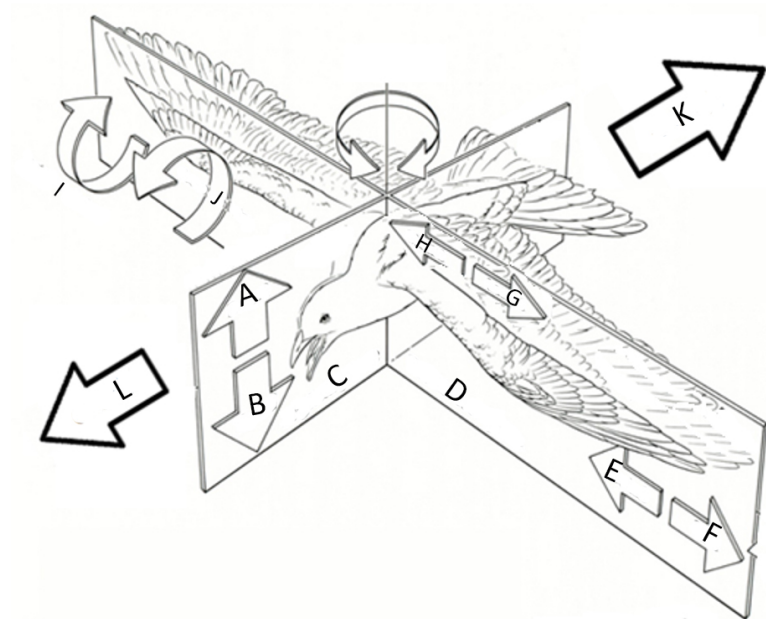


Figure 1. A - Dorsal. B - Ventral. C - Sagittal plane. D - Transverse plane. E - Proximal. F - Distal. G - Lateral. H - Medial. K - Caudal. L - Cranial.

Part I. Dissecting and examining the neck**Dissection tips**

- Dissect the larger diameter end of the neck!
- Cut the soft tissue parts between the vertebrae (e.g. muscles) in a circular way – this makes it easier to reach the joints!

Q.3.3.1 Dissect a cervical vertebrae so that you can examine its surfaces, processes and internal structure as well as its connections with the adjacent vertebrae. Based on your observations, indicate on your answer sheet if each of the following statements is true (T) or false (F).

- A. The connection of the vertebra is characterised by a saddle or sellar joint (*Articulatio sellaris*), in which a concave (hollowed inward) and a convex (domed) surface comprise a joint.
- B. The vertebra is connected with a sledge or gliding joint (*Articulatio delabens*), in which flat joint surfaces glide upon each other.
- C. Regarding the bone connections between vertebrae, the gaps between the precisely fitted surfaces of the bones are filled in by ligaments and fibrous connective tissue (*syndesmosis*), which is visible to the naked eye.
- D. The spinous processes are underdeveloped, therefore the neighbouring vertebrae are not connected by the muscles attaching to the spinous processes.

Q.3.3.2 Examine the intervertebral joints and their movements. Which of the following movements occur at the intervertebral joints? Based on your observations, indicate on your answer sheet if each of the following statements is true (T) or false (F).

- A. Dorsal flexion
- B. Ventral flexion
- C. Lateral flexion
- D. Rotation

Q.3.3.3 Examine the spinal cord in the cervical segment. Based on your observations, indicate on your answer sheet if each of the following statements is true (T) or false (F).

- A. A blood vessel can be found in the midline of the ventral surface of the spinal cord, in the median sagittal plane.
- B. Up to 4 (2 pairs) of spinal nerves can be counted within a 2.5 cm segment of the spinal cord.
- C. The dura mater (outermost membrane of the 3 membranes of the spinal cord) itself cannot be distinguished around the spinal cord, because the dura mater and the periosteum (dense fibrous membrane covering the surface of all bones, in general) of the vertebral bodies are strongly attached together.

Q.3.3.4 Dissect a single vertebra and separate it from the rest of the neck. Based on your observations, indicate on your answer sheet if each of the following statements is true (T) or false (F).

- A. The vertebral body has a sponge-like structure, it is not dense.
- B. No cartilages can be found between the vertebral bodies of chickens.
- C. A pair of bone processes can be observed both on the dorsal and ventral side of the vertebra.
- D. One single vertebral foramen (hole) can be found on the dissected vertebra.

Q.3.3.5 SUBMISSION!

Dissect and prepare a single vertebra and make sure it fulfils the following criteria:

- Whole and intact body
- All processes are intact
- No muscles are attached
- No remnants of the spinal cord

Place the vertebra in the sachet with the code marking. Write this code in the box provided on your answer sheet!

Part II. Dissecting and examining a chicken wing

Anatomy of the bones of the chicken wing

Figure 2 shows the bones of the chicken wing, not necessarily in the order of their direct connections. In the limb, the **upper arm bone** (*humerus*) and the two forearm bones, the **radius** and the **ulna** are significantly elongated. Some of the carpal bones are individual bones, while others are fused with the metacarpals. The unified bone that developed this way has an elongated shape resembling a violin bow. Three fingers (digits I, II and III) are connected to this. The first (I) digit is the thumb, positioned at the side of the radius and consisting of a single phalanx, which bears the bastard wing (***alula***). The second (II), middle digit is the longest with two phalanges, while the third (III) vestigial digit also has one single phalanx, immediately next to II.

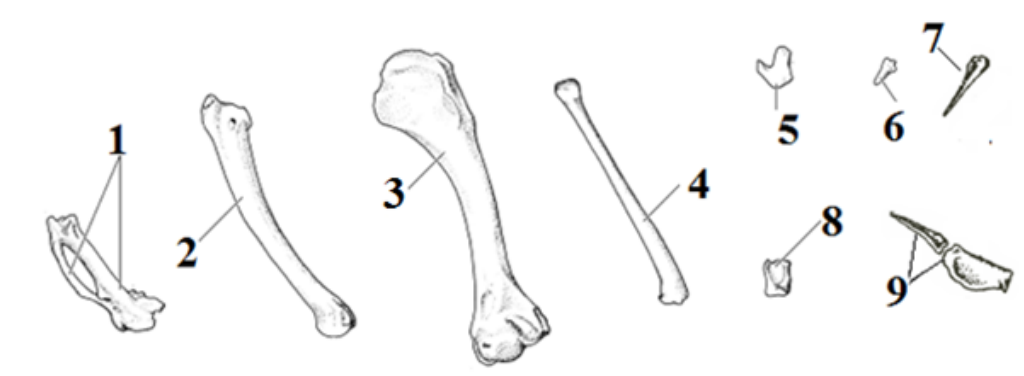


Figure 2. Bones of a chicken wing

Figure 3 demonstrates one of the bones from Figure 2 in more detail. Regarding the anatomical directions and planes, we considered the standard anatomical position as a basis: the wing is open in a lateral position (refer to Figure 1).

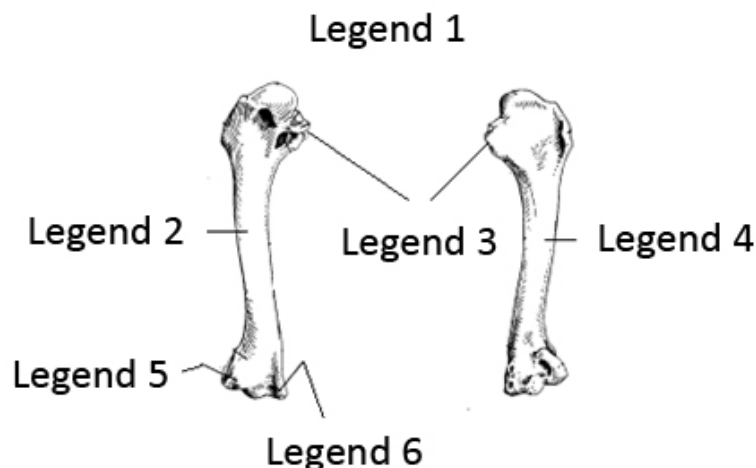


Figure 3. 1 = Proximal end. 2 = Cranial surface. 3 = Tuberculum ventrale. 4 = Caudal surface. 5 = Epicondylus dorsalis. 6 = Epicondylus ventralis.

Dissection tips

- Examine the superficial muscles first and the deeper muscles second, both on the dorsal and ventral sides of the wing.
- Remove the skin from the muscles.
- Open the common connective tissue sheath around the muscles.
- Separate the individual muscles using the tip of closed scissors, breaking the connective tissue between them so that only their own connective tissue sheath (epimysium) is left on them.
- MAKE SURE YOU KEEP ALL MUSCLES INTACT!

Q.3.3.6 Examine how the muscles, bones and joints react to you moving the wing.

Based your observations, indicate on your answer sheet if each of the following statements is true (T) or false (F).

- A. When extending the elbow joint, the bones of the forearm move relative to each other as well: the radius moves in distal, the ulna in proximal direction.
- B. When extending the elbow joint, the bones of the forearm move together, both bones in proximal direction, while during flexion both bones move in distal direction.
- C. When the elbow joint is in flexion, the wrist joint will also bend, thus the metacarpals and digits also move towards the body as the humerus does.
- D. There is a muscle originating from the bottom (distal) end of the humerus and inserting at the top (proximal) end of the *radius*.
- E. The muscle originating from the bottom (distal) end of the humerus and inserting at the top (proximal) end of the *ulna* is a *flexor* muscle.

Q.3.3.7 The three-headed arm muscle, *musculus triceps brachii* extends the elbow joint, while the two-headed arm muscle, *musculus biceps brachii* is the antagonist muscle of the triceps. **Examine the origin and insertion of these muscles. Where are the insertion points of the triceps and the biceps muscles?** Indicate your answer by putting an X in the appropriate box on the answer sheet.

3.3.7.1 *Musculus triceps brachii*

3.3.7.2 *Musculus biceps brachii*

Options:

- A. Radius
- B. Ulna
- C. Both radius and ulna
- D. None of the above

Q.3.3.8 Locate and examine the nerve running along the humerus between the triceps and the biceps on the ventral side of the wing. **Based on what you can see, which of the following muscles does this nerve innervate?** Indicate your answer by putting an X in the appropriate box on the answer sheet.

- A. Biceps
- B. Triceps
- C. Both biceps and triceps
- D. None of the above

Q.3.3.9 SUBMISSION!

Track the nerve running between the triceps and the biceps on the ventral side of the wing, along the humerus, innervating the forearm and also the digits. Excise a single continuous segment that is at least 8 cm long. Make sure you are left with the nerve only!

Place it in the bottle with the code marking. Write this code in the box provided on your answer sheet!

Anatomy of the muscles of the chicken wing

The origin and insertion point of several (not all) wing muscles are listed below. ***Note that m1-m4 are on the ventral side, while m5-m8 are on the dorsal side of the wing.***

m1

- Origin: bottom protuberance of the humerus (ventral epicondyle)
- Insertion: the ulnar side of the wing carpal bone (ulnar carpal bone)

m2

- Origin: humero-carpal ligament and the bottom protuberance of the humerus (ventral epicondyle)
- Insertion: phalanx of digit II

m3

- Origin: proximal third of the body of the ulna
- Insertion: distal phalanx of digit II

m4

- Origin: wing carpal bones, wing metacarpals, space between bones
- Insertion: end phalanx of digit II

m5

- Origin: upper protuberance of the humerus (dorsal epicondyle) *[note: this muscle is positioned at the side of the radius (radial position)]*
- Insertion: wing metacarpals

m6

- Origin: upper protuberance of the humerus (dorsal epicondyle) *[note: this muscle is located next to m8]*
- Insertion: digits I and II

m7

- Origin: from both the radius and ulna
- Insertion: phalanx of digit I

m8

- Origin: upper protuberance of the humerus (dorsal epicondyle) *[note: the outermost muscle attaching to the metacarpals, with dorsal and ulnar position in the forearm]*
- Insertion: wing metacarpals

Q.3.3.10 Examine the muscles on the ventral side of the forearm and the tip of the wing and Figure 4.

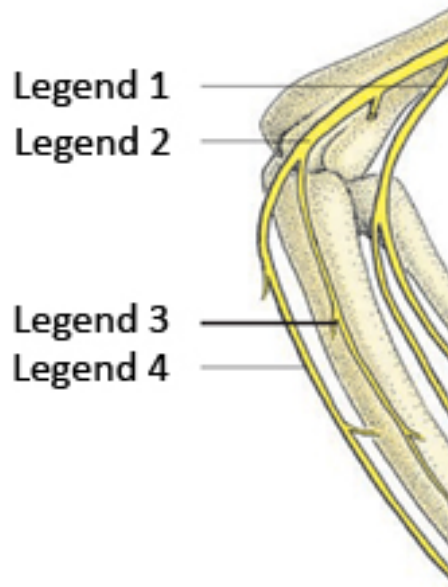


Figure 4. Legend 1 = Median nerve. Legend 2 = Ulnar nerve. Legend 3 = Ramus cranialis. Legend 4 = Ramus caudalis.

Based on your observations, which one of the following statements is true for the innervation of muscles m1 and m3? Indicate your answer by putting an X in the appropriate box on the answer sheet.

- A. from the nerves originating from the upper arm, only the branches of the median nerve innervate both muscles m1 and m3
- B. from the nerves originating from the upper arm, only the branches of the ulnar nerve innervate both muscles m1 and m3
- C. only the ulnar nerve innervate muscle m1, while muscle m3 is innervated only by the median nerve
- D. both muscles (m1 and m3) are innervated by both the ulnar and the median nerve

Q.3.3.11 **Consider the function of muscles m1-m8. Which one of the following pairs comprise synergist muscles?** Indicate your answer by putting an X in the appropriate box on the answer sheet.

- A. m1 and m8
- B. m2 and m7
- C. m2 and m6
- D. m3 and m4
- E. None of the above

- Q.3.3.12** **Examine the muscles at the ventral side of the forearm and the tip of the wing. Identify one muscle in which the length of individual muscle fibres are shorter than the length of the main body of the muscle without tendons.** Indicate your answer by putting an X in the appropriate box on the answer sheet.
- A. m1
 - B. m4
 - C. m5
 - D. m8
 - E. None of the above

- SUBMISSION!**
- Q.3.3.13** The tendon of m7 is also connected to the distal tendon of m5, therefore they have a common tendon.
Excise these two muscles (as a whole) in their entire length, together with the common ligament, from the wing.
Place them in the bottle with the code marking (the same in which you put the nerve segment).
Write this code in the box provided on your answer sheet!

- Q.3.3.14** **Dissect away all the soft tissue so you can observe the bones. How many carpal bones are articulated with the forearm bones?** Indicate your answer by putting an X in the appropriate box on the answer sheet.
- A. The ulna is articulated with two carpal bones, while the radius only with one
 - B. The ulna is articulated with one carpal bone, while the radius with two
 - C. Both the ulna and the radius are articulated with two carpal bones
 - D. Both the ulna and the radius are articulated with one carpal bone
 - E. The ulna is articulated with two carpal bones, while the radius with none

THIS IS THE END OF PRACTICAL PAPER 3.3.

MAKE SURE YOU PUT ALL THE STRUCTURES THAT HAVE TO BE SUBMITTED IN THE CONTAINERS SPECIFIED AND LEAVE THEM ON YOUR TRAY!

MAKE SURE YOU COPY THE CORRECT CODES FROM THESE CONTAINERS INTO THE CORRECT BOXES ON YOUR ANSWER SHEET!

30th International Biology Olympiad SZEGED, HUNGARY



Practical Exam 4.

Bioinformatics

Neurobiology

16th July 2019

COUNTRY

LANGUAGE

Practical Exam 4.

General instructions

This exam consists of two subtasks:

- Subtask 1. Practical Task in Bioinformatics (50 points)
- Subtask 2. Examination of Neuronal Activity (50 points)

1. **Please remember to attach your barcode sticker to all pieces of paper on the answer sheet.**
2. Write your answers in the separate answer sheet provided. **Only answers given in the answer sheet will be considered.**
3. Ensure you received all necessary materials and equipment listed on the next page. If any items are missing indicate this by raising your red card within 10 minutes following the start of the exam.
4. During experiments ensure all materials and equipment is handled properly. Any spilled solutions or broken equipment will not be replaced.
5. Stop answering and put down your pencil immediately when the bell rings signalling the end of the exam.
6. No paper, materials or equipment should be taken out of the laboratory.

IMPORTANT TECHNICAL NOTES ON CHEATING

- **During the practice task, you will use the Geneious Prime and Neurophysiology (Neurofiziológia) software that are already running on your computer.**
- **Please do not use any other software, except for the Geneious and Neurophysiology software that you will use for the completion of the exam.**
- **Your computer records all of your activities, so if you open any other software program, or if you connect your computer to the internet, you will be excluded from this task and lose all points that you scored here.**

Materials and Equipment

- PC running Geneious Prime and Neurophysiology (Neurofiziológia)

SUBTASK 1. PRACTICAL TASK IN BIOINFORMATICS

Modern biological research in the 21st century is inconceivable without bioinformatic analyses. In this practical task, we present an example of how bioinformatic data are processed in order to investigate evolutionary issues of bacterial resistance and sensitivity to antibiotics.

Please do not start using the Geneious Prime software until you have completely read and understood the instructions. Otherwise, you may cause changes to the data that cannot be restored. The lab assistant will not assist you in replacing lost or corrupted data, so you should proceed step by step only.

IMPORTANT TECHNICAL NOTES ON CHEATING

- **During the practical, you will use the Geneious Prime software that is already running on your computer.**
- **Please do not use or even open any other software, except for this Geneious software and the Neurophysiology software that you will use for the other practical that is also already running.**
- **Your computer records all your activities, so if you open any other software, or if you connect your computer to the internet, you will be excluded from this task and lose any points that you scored here.**

Chloramphenicol acetyltransferase (CAT) is responsible for chloramphenicol resistance in *Escherichia coli*. The CAT enzyme is made up of 219 amino acids and encoded by a gene consisting of 660 nucleotides. It is a well-known phenomenon that antibiotic resistance spreads rapidly across pathogens. This is why many scientific studies are looking at ways to prevent this. One approach is to explore whether the inverse of resistance, that is, recovered antibiotic sensitivity, can also be passed on.

In this task, you will be required to analyse the results of such a model experiment. An *E. coli* strain originally resistant to chloramphenicol was exposed to periodic heat stress, and strains were isolated after each stress-inoculation cycle. Some of these strains remained resistant, while others have lost their defences against the antibiotic.

Freshly inoculated liquid cultures of 16 strains from the experiment were grown with chloramphenicol for 2 days. 10 mL of each culture was diluted by 100X. Then we determined the population density of each culture by measuring optical density at 600nm with a spectrophotometer. This value is directly proportional to the population density. Table 1 shows OD_{600} information of the 16 strains.

1.	0.087	5.	0.915	9.	0.869	13.	0.784
2.	0.873	6.	0.852	10.	0.923	14.	0.858
3.	0.110	7.	0.077	11.	0.093	15.	0.102
4.	0.098	8.	0.859	12.	0.920	16.	0.931

When those *E. coli* strain were cultured in medium without Chloramphenicol, no difference in growth was observed among each strain.

Q.4.1.1 Study Table 1 carefully and identify which strain(s) lost its/their resistance. Indicate your answer by putting an X in the appropriate box on the answer sheet.

In order to examine whether strains that became sensitivity have evolved independently from each other or they have a genetic relationship, we will analyse the sequence of a segment of their CAT gene. Since CAT is an enzyme, it can be assumed that its loss of function can be primarily caused by mutations in the active site of CAT.

- Log in to your computer and click on the Geneious Prime icon in the bottom bar:



- cDNA sequence segments of mRNAs from the CAT genes of the examined 16 strains have previously been uploaded.
- You can display them on your screen by clicking on the '+' sign next to the folder entitled 'IBO2019' in the left pane, which will show the folders of this subtask. Clicking on 'Subtask1' will display the sequences.
- Your first task will be to compare the sequences.
- By ticking the boxes to the left of the sequence names, select all 16 sequences, then click on 'Multiple Align' command in the 'Align/Assemble' drop-down menu. In the pop-up window, do not change any settings, and click on OK. The alignment of the sequences can take up to 1 minute, so be patient.
- When the alignment process is completed, the progress bar will disappear and an additional line will appear above the original sequences. Double-click on this new line. You should see a lot of grey bars with some scattered tiny coloured spots on them.
- Zoom the sequence segments with the Ctrl + mouse scroll key combination. The letter identifying the bases will appear. Above the sequences, the consensus sequence is shown which appears to be the most common base sequence during the alignment of the examined instances.
- Use the scroll button of the mouse to move left and right along the sequences.

Answer the following questions in the answer sheet by writing your answer in the appropriate boxes.

Q.4.1.2.a Specify the alignment position in sequence segment 250-350 in which Strain1 has a base different from the consensus sequence. **Write the position number of this base the corresponding box on the answer sheet.**

Q.4.1.2.b Specify the base in Strain1 at the position identified in Q4.1.2.a **by writing the one-letter code of the base in the appropriate box on the answer sheet.**

Q.4.1.2.c Specify the two alignment position(s) where the maximum number of strains showing a difference from the consensus sequence. **Write the alignment positions of these two variants in the corresponding boxes on the answer sheet.**

Q.4.1.2.d Specify the three strains that are likely to carry a deletion mutation in the examined gene segment. **Write the numbers of these strains in the corresponding boxes on the answer sheet.**

Q.4.1.2.e Specify the alignment positions of adjacent-double base changes, wherein both bases were replaced by their complementary ones. **Write the position numbers of these two adjacent bases in the corresponding boxes on the answer sheet.**

Q.4.1.2.f Specify the alignment positions where at least four of the strains specified in your response to question Q4.1.1 have the same changes? **Write each of these position numbers in the corresponding boxes on the answer sheet.**

Your answer to question Q4.1.2.f. is relevant because all the mutations in question can form the basis for the loss of resistance. Your next task will be to check which mutation causes amino acid substitution in the CAT enzyme.

- Close the window where you have worked before (NOT the GENEIOUS program, this window only) by clicking on the 'X' button in the upper right corner.

- Select the 16 DNA sequences by ticking the boxes to their left.
- Click on the 'Translate' submenu under 'Sequence' menu in the menu bar to see the amino acid sequences in the image that appears at the bottom or in a separate window. Similarly to the DNA sequence, the image can be zoomed using the Ctrl+scroll key combination.
- Note that if you zoom into the image to a high magnification, one-letter amino acid codes will be replaced by three-letter codes. In spite of this, ALWAYS USE THE ONE-LETTER AMINO ACID CODES when providing your answers.
- Use the scroll button of the mouse without pressing Ctrl to move left and right along the sequences.

Answer the following questions on the answer sheet.

Q.4.1.3.a

Q.4.1.3.b

Let us suppose that you want to know which amino acid is encoded by the bases 121–123 of the mRNA from the encoding region of the peptide gene in question Q4.1.3.a. Specify the position number of amino acid in the amino acid sequence of the peptide that need to be checked in order to answer the question. **Write the number in the corresponding box on the answer sheet.**

Based on your answers to questions Q4.1.3.a and Q4.1.3.b., check the amino acid positions corresponding to the mutations found in question Q4.1.2.f.

Q.4.1.3.c

Specify the position of the amino acid(s) that changed due to mutations detected in question Q4.1.2.f. **Write this(these) position number(s) in the corresponding boxes on the answer sheet.**

Q.4.1.3.d

Specify the position of the amino acid(s) in which the mutation(s) detected in question Q4.1.2.f. remain(s) silent. **Write this(these) position number(s) in the corresponding boxes on the answer sheet.**

Q.4.1.3.e

Which is the amino acid substitution that appears in all of the chloramphenicol sensitive strains detected in question Q4.1.1.? Specify the position of that amino acid. **Write the position number of this amino acid in the corresponding box on the answer sheet.**

We have identified which amino acid substitutions are likely to cause chloramphenicol sensitivity. Since the examined DNA segment has changed several times and in several loci during the microevolution of the strains, it is also possible to build a DNA-based phylogenetic tree of the 16 strains. You have to do this with the help of Geneious Prime, as specified below.

- If you performed the previous examinations in a new window, close this window (NOT the entire Geneious Prime program!) or if the amino acid sequence appeared in the split bottom pane of the screen, untick the box next to 'Translation of 16 sequences' in the top pane.
- If you can only see the DNA segments again, tick/select the tested 16 DNA sequences, then click on the 'Tree' icon in the top menu bar. In the pop-up window, accept the default settings by clicking on 'OK'. It may take a few seconds for the software to generate the phylogenetic tree. If it appears below only, click the upper right corner of the area displaying the phylogenetic tree to enlarge the image so that you can see the window in full screen.
- The obtained tree is unstructured (unrooted), so you need to transform it so that it can serve as a basis for conclusions. The first step is to select the most divergent strain, which will be the root, of the structured tree.

- Since the original phylogenetic tree does not provide accurate information about the strain most divergent from all the others, you will have to find it in a different way.
- Click on the 'Distances' tab in the second textual menu bar in the upper left. Thus, a distance matrix is obtained that shows the degree of divergence between each pair of DNA sequences. In the upper middle menu, change 'Styles' to 'Heatmap'. The darkness of rectangles corresponds to the extent of divergence.

Q.4.1.4.a Specify the number of the strain that probably has the most divergent sequence among the 16 strains. **Write this strain number in the corresponding box on the answer sheet.** (For example, if your answer is Strain-23, write '23' on answer sheet.)

Based on the above data, you can already start structuring the phylogenetic tree.

- To do this, switch back to the 'Tree view' view by clicking on the appropriate tab in the second menu bar from the top.
- Subsequently, click on the endpoint nearest to the name of the strain identified in question Q4.1.4.a., and then click on the 'Root' button in the third menu bar that is activated at this time. Thus, an already structured tree is obtained where the common ancestor is the strain identified in question Q4.1.4.a.
- For a simpler view, tick the 'Transform branches' box in the right menu bar (Formatting submenu)
- Change the scaling from "Equal" to "Cladogram" using the scroll down menu just below.

Q.4.1.4.b Find the branch containing the sensitive mutants and specify the strain-number of the resistant strain(s) that is most closely related to these strains. **Write these numbers in the corresponding boxes on the answer sheet.** (For example, if your answer is Strain-23, write '23' on the answer sheet.)

Q.4.1.4.c Specify the strain-number of the two sensitive strains that may have evolved most recently. **Write these numbers in the corresponding boxes on the answer sheet.** (For example, if your answer is Strain-23, write '23' on the answer sheet.)

Q.4.1.4.d For each of the following groups of strains, specify whether it is monophyletic (A), or NON-monophyletic (B). **Indicate your answer by putting an X in the appropriate box on the answer sheet.**

Groups:

I. 8; 12

II. 5; 6; 16

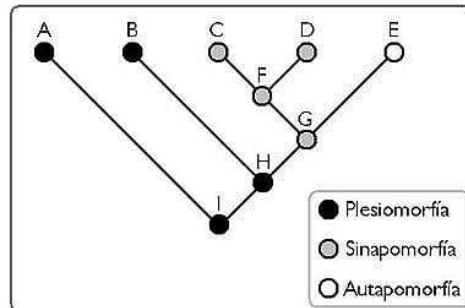
III. 3; 7; 10

Options:

A – monophyletic

B – NON-monophyletic

Q.4.1.4.e Based on the phylogenetic tree obtained above, you can find autapomorphic, synapomorphic, and plesiomorphic mutations in certain nucleotide positions in the sequences examined in the first part.



Return to the 16-line DNA sequence alignment obtained in the first part by double-clicking on the 'Nucleotide alignment' line and specify the types of mutations in the following positions from the perspective of the phylogenetic tree. **Indicate your answer by putting an X in the appropriate box on the answer sheet.**

Positions:

51
144
312
447
531

Options:

A – autapomorphic
B – plesiomorphic
C – synapomorphic

SUBTASK 2. EXAMINATION OF NEURONAL ACTIVITY PATTERN ON MOVING ANIMALS

IMPORTANT TECHNICAL NOTES ON CHEATING

- During the practice task, you will use the **Neurophysiology (Neurofiziológia)** software that is already running on your computer.
- Please do not use any other software, except for the **Geneious** software that you will use for the other bioinformatics task.
- Your computer records all of your activities, so if you open any other software program, or if you connect your computer to the internet, you will be excluded from this task and lose all points that you scored here.

Please, do not start the software yet. Read the detailed descriptions on the next pages first.

Your task is to clarify the role of neurons involved in navigation. Navigating space is a complex problem involving many neurons. The 2014 Nobel prize was awarded to O'Keefe, and the 2011 Brain prize was awarded to the Hungarian Gyorgy Buzsaki, for developing techniques to study this problem. We can observe the individual activity of hundreds of neurons. Multiple electrodes are inserted into rodent brains. After the surgery, the animal recovers and adapts to the device on its head. Next, the animal is placed into experimental situations, and measurements are performed (Figure 1).

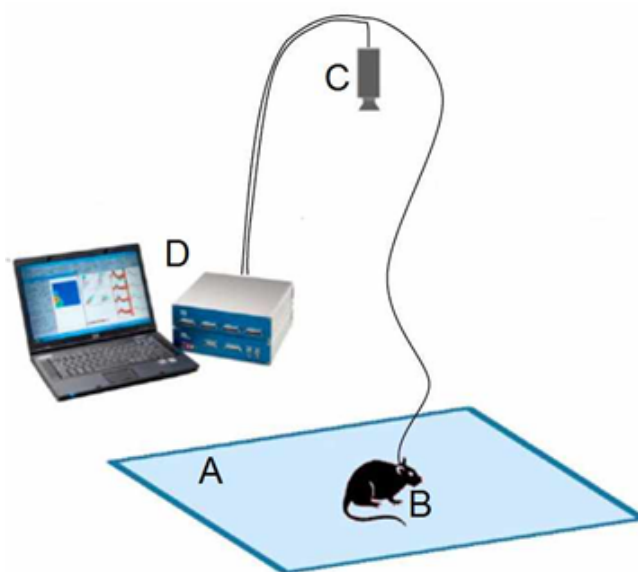


Figure 1. Experimental set-up. **A:** Exploration field. **B:** Animal with electrodes after surgery. **C:** Camera recording the path of the animal. **D:** Recording unit (records changes in potential of brain regions and animal movements simultaneously)

Using similar techniques shown above, O'Keefe investigated the hippocampus and found that many of its neurons fire when the animal is standing on or passing through a given point of its exploration field. These cells were named *place cells*. The results of such an experiment are shown on Figure 2.

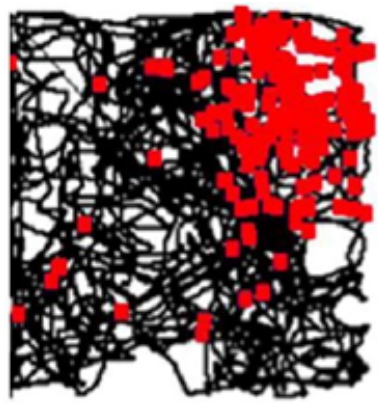


Figure 2. Place cell behaviour. **Black line:** The path a rodent took as it explored a square field. **Red dots:** Position of the rodent where a specific neuron fired

The place cells described by O'Keefe are not measured in your data! They are not a cell-type you should look for!

Summary of Tasks

Your task is to analyse the function of neurons in the entorhinal cortex. The task will consist of two parts:

- **Part 1.** You will have to analyse recordings of 30 cells. You will identify which neurons are responsible for perceiving the same aspects of movement and/or position in space. At first, you do not have to state which aspects of movement or position each neuron responds to, you only have to identify groups of cells performing similar functions.
- **You may only begin the second part after completing the first part.**
- **Part 2.** When beginning the second part, the lab assistant will provide you additional sheets with a short description of the exact function of the cells in part 1. After studying these descriptions, you will have to state the precise aspects of movement/space which six neurons respond to.

The background of your experiment:

1. A rat implanted with electrodes was allowed to explore a 2m x 2m square field.
2. We recorded the animal's movement and the activity of different neurons.
3. Results are displayed in the Neurophysiology software.
4. You can switch between the NEUROPHYSIOLOGY and the Geneious Prime softwares by clicking on the system tray or using the Alt+Tab key combination.
5. **Do not use the software before reading the tasks.**

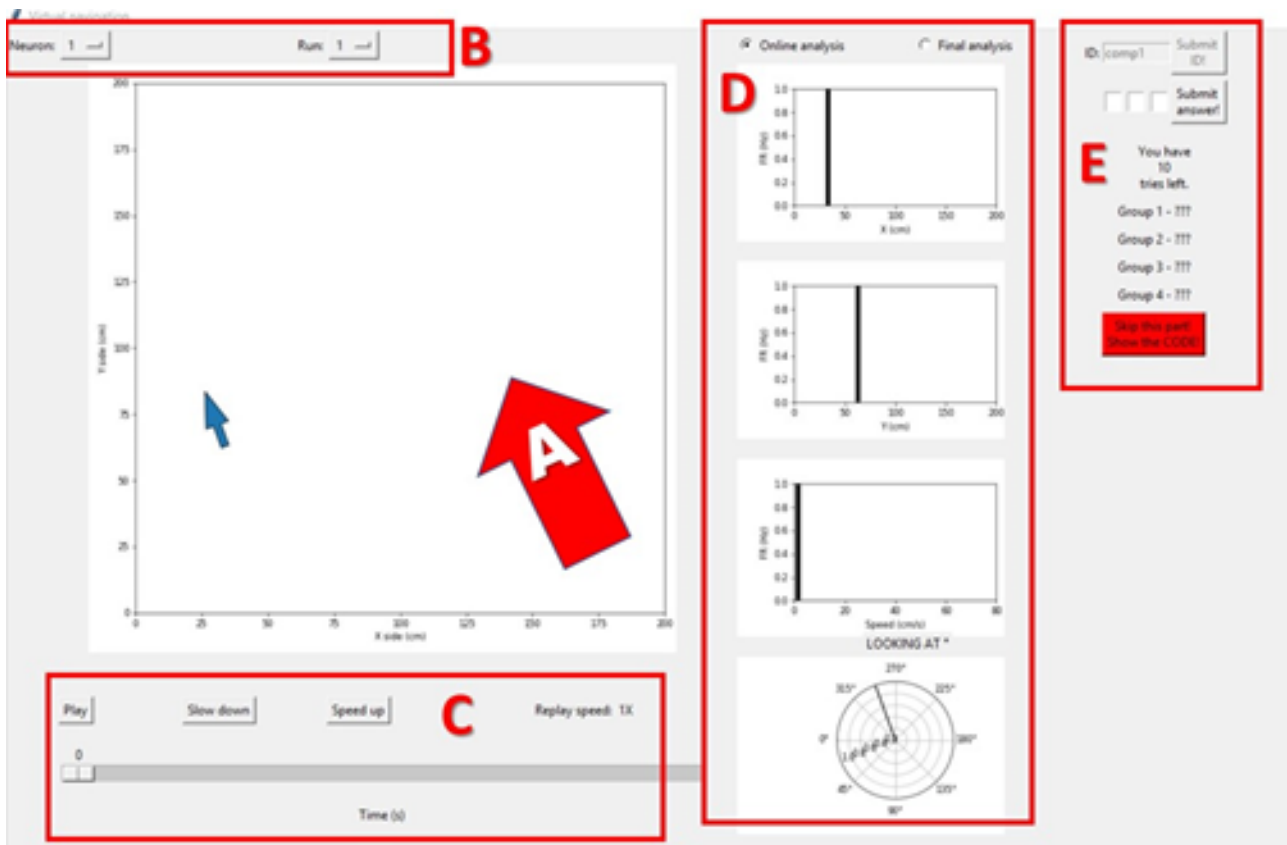


Figure 3. Software overview. A: Exploration field. B: Selection of neuron and run. C: Playback control. D: Statistical information on the actual run. E: Answer block

Part 1

Your software displays the route of the experimental animal continuously in time. You are able to select individual neurons, and red dots appear on the animal's route when the firing of this neuron changes significantly. Each animal was recorded doing three separate runs, so three data sets are available for each neuron. (Figure 4-6).

In Part 1, the animal's route will fade with time, so you can only observe a portion of the rat's movements.

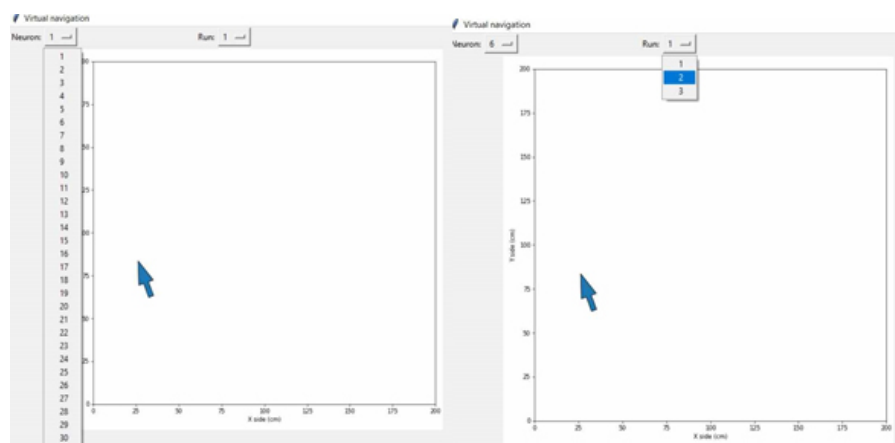


Figure 4. Selecting neurons and runs. In this menu, you can switch between individual neurons ("neurons" 1-30). Click on "run" and choose 1-3 from the drop-down menu.

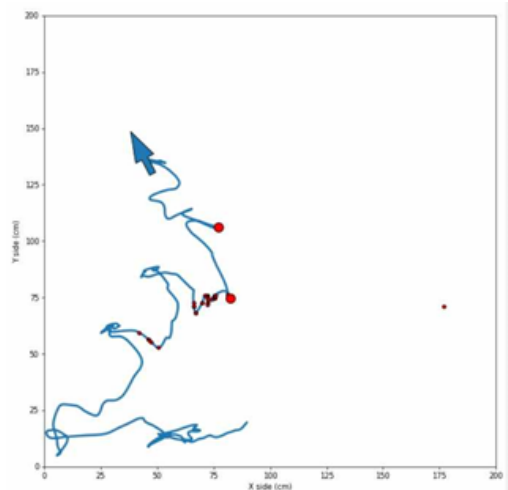


Figure 5. Exploration field. Blue shows route of animal, red dots show neuron activity and the arrow shows the direction of the head.

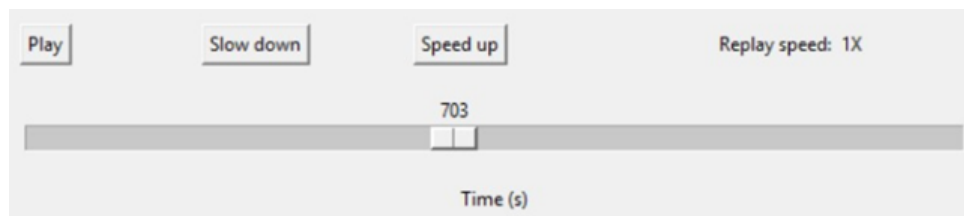


Figure 6. Playback control. You can start ('Play') or pause ('Pause') the run, or you can speed up ('Speed Up') or slow down ('Slow Down') the playback. Using the slider, you can jump forward or backward in a run. **Please be patient when using the slider, the software may take a few seconds to load.**

You will see aggregated data of the activity of the neuron (y-axis) plotted against aspects of its movement (Figure 7).

'Online Analysis': the graphs show aggregated data collected until the current time point of the current playback. The rat's current position is represented by black lines on the graphs.

'Final Analysis': the graphs show aggregated values of the entire current run. The black line returns to the 0 positions.

FR(Hz) vs. X (cm): the firing frequency of the neuron is plotted against the animal's horizontal position measured from the lower left corner of the exploration field.

FR(Hz) vs. Y (cm): the firing frequency of the neuron is plotted against the animal's vertical position measured from the lower left corner of the exploration field.

FR (Hz) vs. Speed (cm/s): the firing frequency of the neuron is plotted against the animal's linear speed.

LOOKING AT: the firing frequency of the neuron mapped against the angle between the animal's direction of gaze and the straight line connecting the center of the exploration field with its lower left corner.

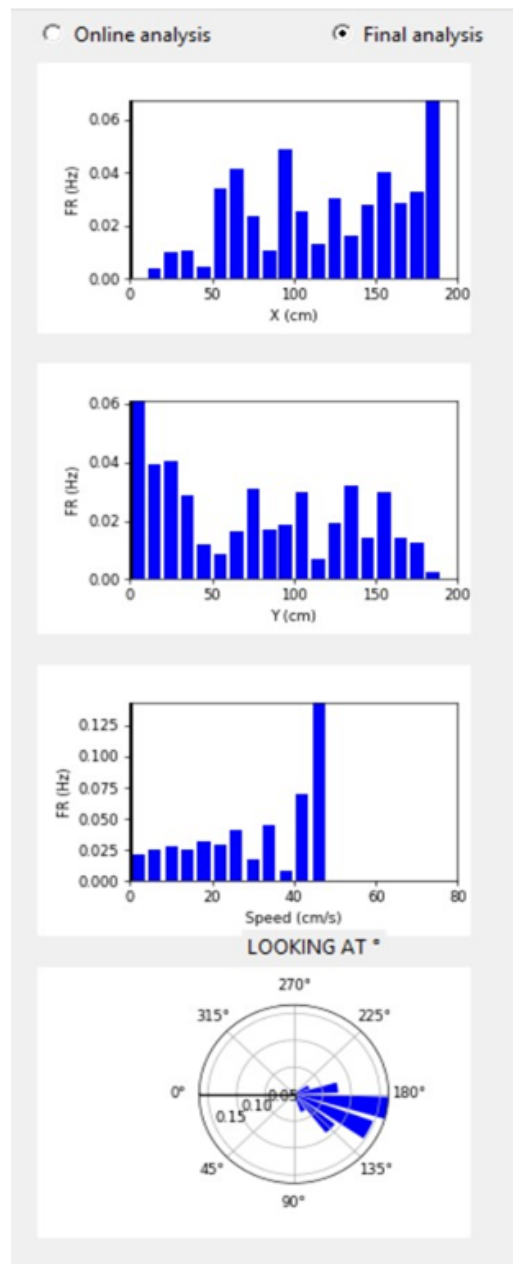


Figure 7. Statistical information on the run

4.2.1.Q You will have to use the exploration field and statistical information to determine whether each neuron has a function in the perception of movement and position characteristics. You will then have to find cells with similar function.

- When you have an idea what the function of a certain neuron is, use the software to look for cells with similar function.
- **The cells measured in your data are divided into FOUR groups based on the different characteristics of movement and position they respond to.**
- **Each cell only responds to one of these four aspects of movement and position.**
- **Approximately half of the 30 cells measured in your data do not respond to movement or position. These are NOT cells which you are interested in.**

- Your data includes at least FOUR neurons from each group.
- **You must find THREE neurons from each group**
 - I.e. you must identify four groups of three neurons
 - All the neurons in each group you enter should have the same function
 - All the groups you enter should have different functions

You have a maximum of TEN ATTEMPTS at entering groups of three neurons to find all four categories

- **You will receive fewer marks for each attempt**
 - I.e. You should aim to find all four groups in your first four attempts
 - You can use table 1 to keep track of the attempts you make. Your notes on paper will NOT be marked.

T1			
T2			
T3			
T4			
T5			
T6			
T7			
T8			
T9			
T10			

Table 1

When you identify three neurons from the same group, enter them into the answer block (Figure 8). Repeat for each group.

Figure 8. Answer block

Write the numbers of three neurons into the rectangles, then click “Submit answer”. The system counts down your attempts from 10 (“You have 10 tries left”)

If your answer is correct you will see this box (Figure 9). Click “OK” and continue to work.

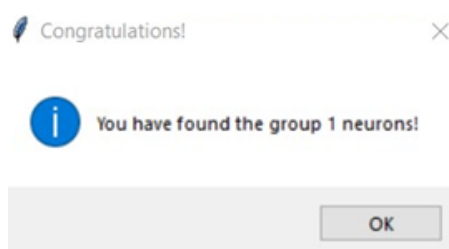


Figure 9. Correct answer

If your answer is correct, but the neurons are in a group that you have already correctly submitted, then you will see this box (Figure 10). This will count as a false attempt and the number of remaining attempts will decrease. Click “OK” and continue to work.

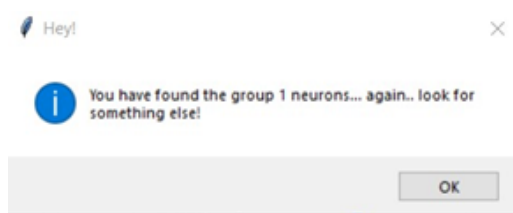


Figure 10. Re-discovery of a group

If you submit three neurons which are not in the same group, you will not see a pop-up box, but the number of remaining attempts will decrease, and this will count as a false attempt. Continue to work.

When you correctly identify three neurons in each of the four groups, part 1 will end. Or, when you run-out of attempts, part 1 will end. You will see this box (Figure 11).

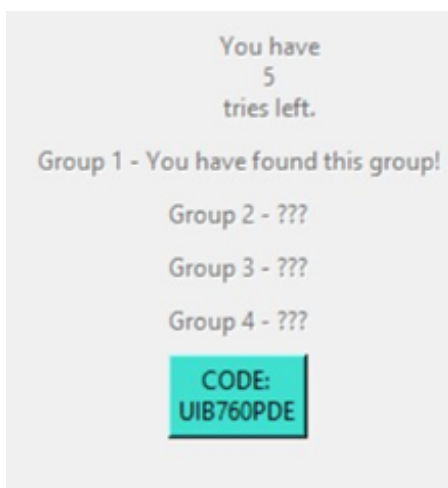


Figure 11. Completion of Part 1.

You MUST write the "CODE: #####" on your ANSWER SHEET.

Alternatively, you **may choose to end Part 1** before you run out of attempts to save time. Click "Skip this part! Show the CODE!" This box will appear (Figure 12). Click "YES" to end the task and reveal your code. This is irreversible, you will receive marks for the correct groups you have discovered but it will count as using all ten attempts. You must write your code on your answer sheet. Click "NO" to return to part 1 and continue working.



Figure 12. Giving-up

After completing part 1 you MUST hold up your YELLOW CARD.
An assistant will check that you have correctly written the code.
An assistant will give you INFORMATION necessary for PART 2.

Part 2

Read the information given to you by the assistant describing five different groups of neurons (A, B, C, D & E). Your software will now display the whole route of the animal without fading (Figure 13).

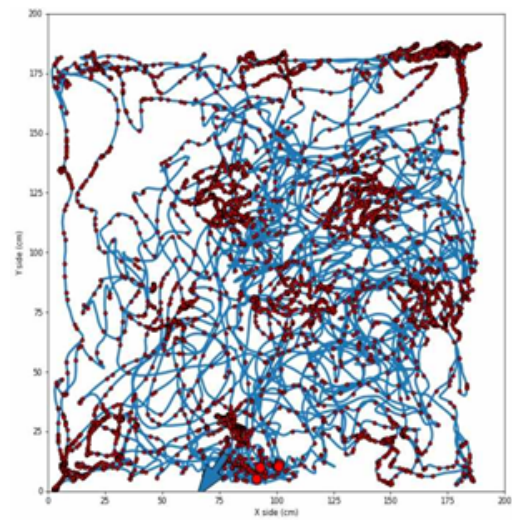


Figure 13. Exploration field for Part 2.

Six specific neurons are listed on your answer sheet (Number 23, 17, 9, 25, 3 & 11). View the Exploration fields of runs 1-3 for each of these neurons. Identify which of the five different groups each neuron belongs to.

Q.4.2.2 Put an X on your ANSWER SHEET identifying which GROUP EACH NEURON belongs to.

Extra information on neuron types

A) Speed cells

Speed cells' firing rate is proportional to the linear speed of the animal

Researchers recorded neuronal activity whilst controlling rats' running speed. Rats were put in a bottomless frame which was moved at pre-set speeds, compelling them to run (Fig. 1a). Rats either ran with gradually increasing speed (Figure 1b left, middle), or with a sharp transition in the middle (Figure 1b right).

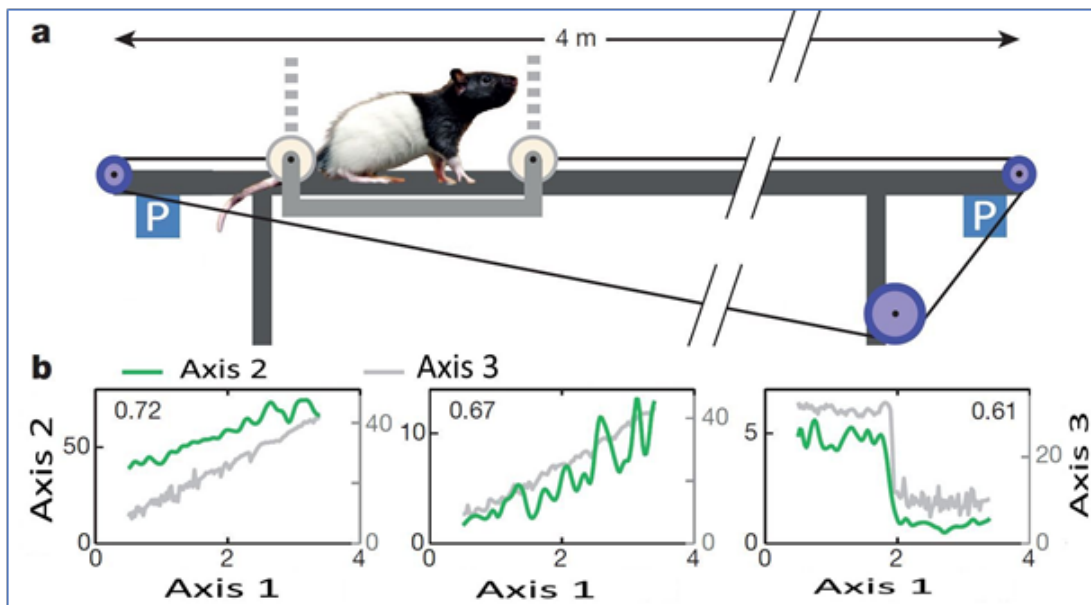


Figure 1. Speed set-up. Axis 1: Position in track (m). Axis 2: Firing Rate (Hz). Axis 3: Speed (cm/s)

The other cell-types were discovered with the experimental design shown in Part 1.
Results shown in Figure 2.

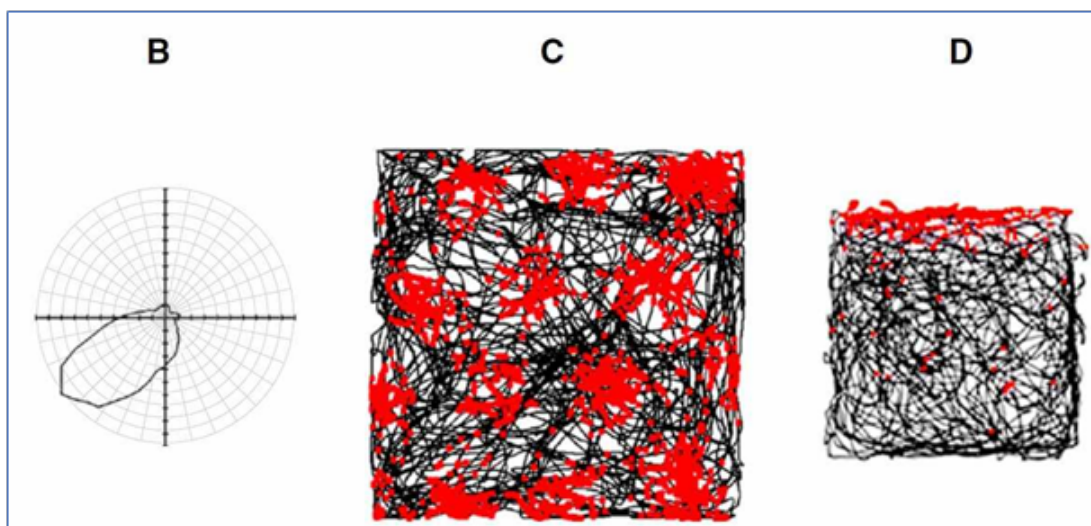


Figure 2

B) Head direction cells

The diagram depicts the firing rate (distance from origin) as a function of the animal's head direction. Head direction cells fire anywhere in an environment when the animal is looking in a certain direction.

C) Grid cells

Grid cells have firing areas evenly spaced in a close-packed hexagonal array across the surface of the field.

D) Border cells

Border cells only fire when the rat is in close proximity to the wall of the field.

E) Bulk cells

These cells do not show a clear firing pattern when the animal is moving even though they are often firing during locomotion. This means we do not understand their function yet.